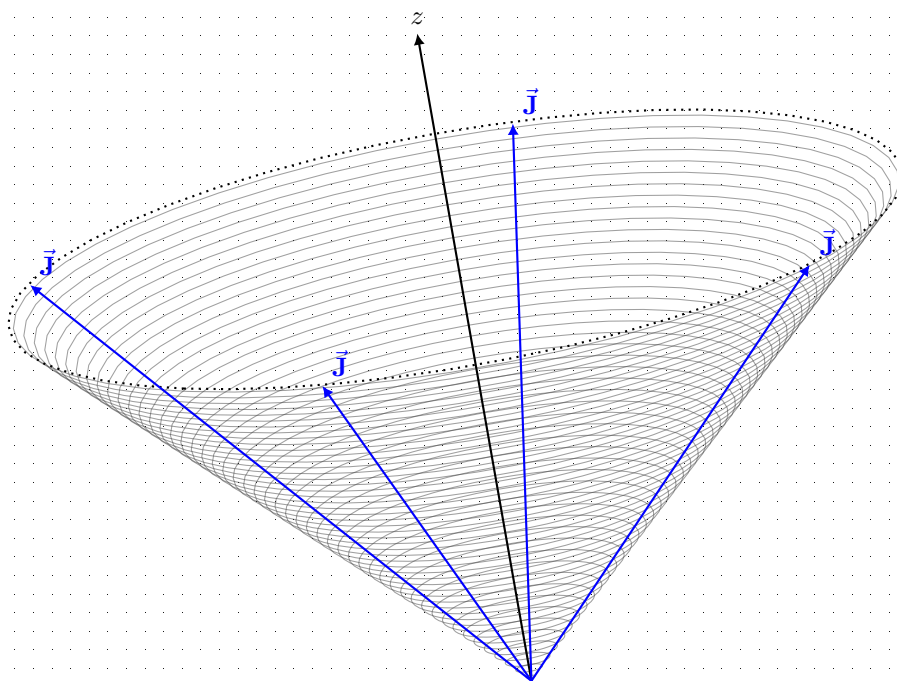


Lecture Notes on Angular Momentum Algebra

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Dedication

Dedicated to my Father,

Everything I am, I owe to you. I will see you in the stars.

Preface

As I think of “Understanding a topic”, it cannot be explained in a purely classical sense: either you understand something completely or you simply do not. There is no true intermediate state, no superposition.

When, in 1913, “Bohr” proposed his model for the “simple hydrogen atom”, one of the first things he introduced was the “quantization of angular momentum”. This idea allowed the atom to possess discrete energy states rather than a continuous spectrum.

In a similar spirit, the same principle applies to learning any subject. One gradually moves from confusion to clarity, step by step, until the concepts become well defined and discrete in the mind.

These notes are intended to guide you from the basics to more advanced topics, building the ideas carefully and systematically. I hope they help you develop a clear and solid understanding of the subject.

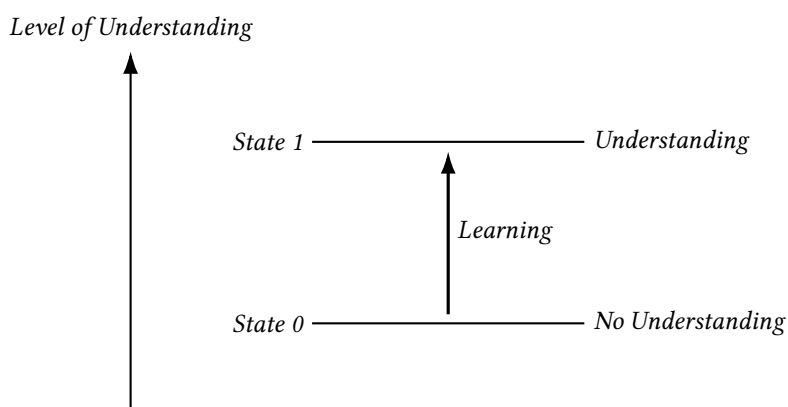


Figure 1: Two-level picture of understanding.

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All you need to know about Angular Momentum

1.1 Introduction

Till now we have studied how to solve the “Time-Dependent Schrödinger Equation” only for the one dimensional case, which is rather elementary. However, if we want to study realistic physical systems, we must move on to the three dimensional case.

Before proceeding to the 3D case, it is essential to understand the concept of angular momentum in quantum mechanics. As we will see later on, in three dimensions the motion of a particle can be separated into radial and angular parts, and the angular dependence of the “wavefunction” is governed by the angular momentum operators. The eigenfunctions of these operators are the “spherical harmonics”, which naturally appear when solving the “Schrödinger equation in spherical coordinates”. Therefore, a proper understanding of angular momentum is crucial for solving many important three dimensional quantum systems such as atoms, molecules, and central potential problems.

Angular momentum plays a fundamental role in quantum mechanics. Many physical systems possess rotational symmetry, and the corresponding conserved quantity is the angular momentum. Understanding the properties of angular momentum operators and their eigenvalues allows us to analyse a wide range of physical phenomena.

One important application appears in the study of rotational motion of molecules. In diatomic and polyatomic molecules the rotational degrees of freedom lead to discrete rotational energy levels. Transitions between these levels produce characteristic absorption or emission lines in molecular spectra. By analysing these rotational spectra, important information about molecular structure, such as the moment of inertia and bond lengths, can be obtained.

Another important application is the “Stern-Gerlach experiment”, which demonstrated the quantization of angular momentum. When a beam of atoms passes through an inhomogeneous magnetic field, the beam splits into discrete components rather than forming a continuous distribution. This result provided direct experimental evidence of space quantization and revealed the existence of intrinsic angular momentum, or spin.

Thus, the study of angular momentum is essential for understanding atomic and molecular structure, spectroscopy, and many fundamental quantum phenomena.

1.2 Orbital Angular Momentum

For a particle with momentum $\vec{p}$ and position $\vec{r}$, the angular momentum is given by

$$\vec{L} = \vec{r} \times \vec{p} \quad (1.1a)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} \quad (1.1b)$$

$$= (yp_z - zp_y)\hat{i} + (zp_x - xp_z)\hat{j} + (xp_y - yp_x)\hat{k} \quad (1.1c)$$

Thus in component form,

$$L_x = yp_z - zp_y, \quad L_y = zp_x - xp_z, \quad L_z = xp_y - yp_x \quad (1.2)$$

Note that $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are unit vectors, not operators. The corresponding quantum operators are obtained by replacing $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$ with their operator forms:

$$\vec{\mathbf{r}} \rightarrow \hat{\mathbf{r}} = (\hat{x}, \hat{y}, \hat{z}), \quad (1.3)$$

$$\vec{\mathbf{p}} \rightarrow \hat{\mathbf{p}} = -i\hbar \nabla \quad (1.4)$$

Explicitly,

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \quad \hat{p}_y = -i\hbar \frac{\partial}{\partial y}, \quad \hat{p}_z = -i\hbar \frac{\partial}{\partial z} \quad (1.5)$$

Therefore,

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = -i\hbar \hat{\mathbf{r}} \times \nabla \quad (1.6)$$

Now Cartesian Components of $\hat{\mathbf{L}}$ are written as :

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \quad (1.7a)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \quad (1.7b)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \quad (1.7c)$$

Classically the angular momentum square is given as :

$$\vec{\mathbf{L}} \cdot \vec{\mathbf{L}} = L_x^2 + L_y^2 + L_z^2 \quad (1.8)$$

Similar way the corresponding quantum mechanical version is defined as :

$$\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \quad (1.9)$$

1.2.1 Commutation Algebra

$\hat{x}, \hat{y}$ and $\hat{z}$ mutually commute and so does $\hat{p}_x, \hat{p}_y$ and $\hat{p}_z$. But we have :

$$[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar \quad (1.10)$$

¹Note that all these operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ and $\hat{\mathbf{L}}^2$ are hermitian. This means they represent physical observables and they have real eigenvalue.

Using these we now try to find the following commutation :

$$\begin{aligned}
 [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] \\
 &= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] \\
 &= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z, \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z] \\
 &= \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x - 0 - 0 + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\
 &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y \\
 &= i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) \\
 &= i\hbar\hat{L}_z
 \end{aligned}$$

Similarly we can calculate the rest of the commutation, but we can get them immediately by doing a cyclic permutation of indices i.e. $x \rightarrow y, y \rightarrow z, z \rightarrow x$. Finally we have :

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x, \quad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y \quad (1.11)$$

Problem 1.1

Find out the following commutators

1. $[\hat{L}_z, \hat{\mathbf{r}}^2]$
2. $[\hat{L}_z, \hat{\mathbf{p}}^2]$

Where $\hat{\mathbf{r}}^2 = \hat{x}^2 + \hat{y}^2 + \hat{z}^2$ and $\hat{\mathbf{p}}^2 = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2$.

Problem 1.2 “Ladder Operators”

We define the “Ladder Operators” as follows :

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y \quad (1.12a)$$

$$\hat{R}_{\pm} = \hat{x} \pm i\hat{y} \quad (1.12b)$$

Prove that :

1. $[\hat{L}_{\pm}, \hat{R}_{\pm}] = \pm 2\hbar\hat{z}$
2. $[\hat{L}_{\pm}, \hat{R}_{\mp}] = 0$
3. $[\hat{L}_{\pm}, \hat{z}] = \mp \hbar\hat{R}_{\pm}$
4. $[\hat{L}_z, \hat{R}_{\pm}] = \pm \hbar\hat{R}_{\pm}$

1.2.2 Consequence of the commutation

Directly inspecting Eq. (1.11) we see that the operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ do not commute themselves. What this means is that “quantum mechanically” if we want to measure the components of “angular momentum” for a particle or a body, we can not measure them simultaneously to arbitrary accuracy. This is completely an opposing statement from the point of

view of “classical mechanics”, where this is always possible. According to the generalized uncertainty principle we have :

$$\begin{aligned}\sigma_{L_x} \sigma_{L_y} &\geq \left| \frac{1}{2i} \langle [\hat{L}_x, \hat{L}_y] \rangle \right|^2 \\ &\geq \left| \frac{1}{2i} \langle i\hbar \hat{L}_z \rangle \right| \\ &\geq \frac{\hbar}{2} |\langle \hat{L}_z \rangle|\end{aligned}$$

What it tells us that suppose if we want to measure “ x ” and “ y ” components of the angular momentum simultaneously, then we are bound by this relation.

1.2.3 Short Hand notation of commutation algebra

In short hand notation, the commutation relations described in Eq. (1.11) can be written as :

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k \quad (1.13)$$

Here ϵ_{ijk} is “levi-civita” symbol, which is completely “anti-symmetric” in its indices. The defining thing is that $\epsilon_{ijk} = +1$ for cyclic permutations of ijk . Our job would be to prove it which is done in the chapter A

Problem 1.3

Evaluate the following :

- (a) $[\hat{\mathbf{L}}^2, \hat{L}_x]$
- (b) $[\hat{\mathbf{L}}^2, \hat{L}_y]$
- (c) $[\hat{\mathbf{L}}^2, \hat{L}_z]$

1.2.4 Finding simultaneous eigenstates

Once you are done solving **Problem (1.3)** you will find that all the commutations are zero. In a more compact notation we can write :

$$[\hat{\mathbf{L}}^2, \hat{\mathbf{L}}] = 0^3 \quad (1.14)$$

Now if two operators commute they can simultaneous eigenstates and we can measure them simultaneously with arbitrary precision. So $\hat{\mathbf{L}}^2$ commutes with $\hat{L}_x$ and also with $\hat{L}_y$ and $\hat{L}_z$. But you see the operators $\hat{L}_x$, $\hat{L}_y$ and $\hat{L}_z$ do not commute with each other. So we take one component of the angular momentum (commonly) $\hat{L}_z$ and try to find out the simultaneous eigenstates of the operators $\hat{\mathbf{L}}^2$ and $\hat{L}_z$ and also the corresponding eigenvalues. Which is basically done throughout the following sections Secs. 1.4 and 1.8.

1.3 Angular Momentum Operators in Spherical Polar Coordinates

The angular momentum operator is given by :

$$\hat{\mathbf{L}} = -i\hbar (\hat{\mathbf{r}} \times \nabla) \quad (1.15)$$

²For two observables A, B generalized uncertainty principle is given by : $\sigma_A \sigma_B \geq \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right|$ with $\sigma_A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}$.

³Here $\hat{\mathbf{L}} = \hat{L}_x \hat{\mathbf{i}} + \hat{L}_y \hat{\mathbf{j}} + \hat{L}_z \hat{\mathbf{k}}$

Now the gradient operator(∇) in spherical polar coordinates is given by :

$$\nabla = \hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \frac{\hat{\mathbf{e}}_\theta}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\mathbf{e}}_\phi}{r \sin \theta} \frac{\partial}{\partial \phi} \quad (1.16)$$

Please note that, $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi$ are the unit vectors, not the quantum mechanical operators. Using $\hat{\mathbf{r}} = r\hat{\mathbf{e}}_r$ we have,

$$\begin{aligned} \hat{\mathbf{r}} \times \nabla &= \begin{vmatrix} \hat{\mathbf{e}}_r & \hat{\mathbf{e}}_\theta & \hat{\mathbf{e}}_\phi \\ r & 0 & 0 \\ \frac{\partial}{\partial r} & \frac{1}{r} \frac{\partial}{\partial \theta} & \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{vmatrix} \\ &= -\hat{\mathbf{e}}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} + \hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \end{aligned}$$

Which gives :

$$\hat{\mathbf{L}} = -i\hbar \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \hat{\mathbf{e}}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) \quad (1.17)$$

Expressing the unit vectors $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\phi$ in terms of Cartesian unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ we have :

$$\hat{\mathbf{e}}_\theta = \cos \theta \cos \phi \hat{\mathbf{i}} + \cos \theta \sin \phi \hat{\mathbf{j}} - \sin \theta \hat{\mathbf{k}} \quad (1.18a)$$

$$\hat{\mathbf{e}}_\phi = -\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}} \quad (1.18b)$$

$$\begin{aligned} \hat{\mathbf{L}} &= -i\hbar \left[(-\sin \phi \hat{\mathbf{i}} + \cos \phi \hat{\mathbf{j}}) \frac{\partial}{\partial \theta} - (\cos \theta \cos \phi \hat{\mathbf{i}} + \cos \theta \sin \phi \hat{\mathbf{j}} - \sin \theta \hat{\mathbf{k}}) \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) \right] \\ &= -i\hbar \left[\hat{\mathbf{i}} \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) + \hat{\mathbf{j}} \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) + \hat{\mathbf{k}} \frac{\partial}{\partial \phi} \right] \\ &= \hat{L}_x \hat{\mathbf{i}} + \hat{L}_y \hat{\mathbf{j}} + \hat{L}_z \hat{\mathbf{k}} \end{aligned}$$

Comparing the components we have :

$$\hat{L}_x = -i\hbar \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \quad (1.19a)$$

$$\hat{L}_y = -i\hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \quad (1.19b)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \quad (1.19c)$$

For the ladder operators :

$$\begin{aligned} \hat{L}_+ &= \hat{L}_x + i\hat{L}_y \\ &= -i\hbar \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) + i(-i\hbar) \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \\ &= -i\hbar \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) + \hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \\ &= [i\hbar \sin \phi \pm \hbar \cos \phi] \frac{\partial}{\partial \theta} + \cot \theta [i\hbar \cos \phi \mp \hbar \sin \phi] \frac{\partial}{\partial \phi} \\ &= \pm \hbar [\cos \phi \pm i \sin \phi] \frac{\partial}{\partial \theta} + i\hbar \cot \theta [\cos \phi \pm i \sin \phi] \frac{\partial}{\partial \phi} \\ &= \pm \hbar e^{\pm i\phi} \frac{\partial}{\partial \theta} + i\hbar \cot \theta e^{\pm i\phi} \frac{\partial}{\partial \phi} \quad 4 \end{aligned}$$

We finally obtain :

$$\hat{L}_{\pm} = \pm \hbar e^{\pm i\phi} \left[\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right] \quad (1.20)$$

Problem 1.4

Find out the form of the following operators :

(a) $\hat{L}_+ \hat{L}_-$

(b) $\hat{L}_- \hat{L}_+$

Take any test function $f(\theta, \phi)$ and operate on them, using $\hat{L}_{\pm}$ forms given by Eq. (1.20).

Example 1.1 Finding out the form of $\hat{L}^2$

Once you do **Problem (1.4)** you will find :

$$\hat{L}_+ \hat{L}_- = -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \cot^2 \theta \frac{\partial^2}{\partial \phi^2} + i \frac{\partial}{\partial \phi} \right] \quad (1.21)$$

Later on you will find in Example 2.3 :

$$\hat{L}^2 = \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ + \hbar \hat{L}_z \quad (1.22)$$

Taking the upper sign we have :

$$\begin{aligned} \hat{L}^2 &= \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ + \hbar \hat{L}_z \\ &= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \cot^2 \theta \frac{\partial^2}{\partial \phi^2} + i \frac{\partial}{\partial \phi} + \frac{\partial^2}{\partial \phi^2} - i \frac{\partial}{\partial \phi} \right]^5 \\ &= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + (\cot^2 \theta + 1) \frac{\partial^2}{\partial \phi^2} \right] \\ &= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \csc^2 \theta \frac{\partial^2}{\partial \phi^2} \right] \\ &= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\ &= -\hbar^2 \left[\frac{\sin \theta}{\sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\partial}{\partial \theta} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \right) \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\ &= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \end{aligned} \quad (1.23)$$

Problem 1.5 Finding out the form of $\hat{L}^2$ “Other Way round”

Using the form of $\hat{L}_- \hat{L}_+$ obtain the form of $\hat{L}^2$ using $\hat{L}^2 = \hat{L}_- \hat{L}_+ + \hat{L}_+ \hat{L}_- + \hbar \hat{L}_z$, the same as we did before.

Hint: We can use the commutator $[\hat{L}_+, \hat{L}_-] = 2\hbar \hat{L}_z$ to obtain $\hat{L}_- \hat{L}_+$ given that we know $\hat{L}_+ \hat{L}_-$.

Compare the results you have obtained this way and the way you did before. (They should be the same).

⁴Where we have used the Euler's formula $e^{\pm i\phi} = \cos \phi \pm i \sin \phi$.

⁵We have used the fact that $\hat{L}_z^2 = (-i\hbar)^2 \frac{\partial^2}{\partial \phi^2} = -\hbar^2 \frac{\partial^2}{\partial \phi^2}$

Problem 1.6

Find out the form of the following operators :

(a) $[\hat{L}_+, \hat{L}_-, \hat{L}_- \hat{L}_+]$

(b) $\{\hat{L}_+, \hat{L}_-, \hat{L}_- \hat{L}_+\}$

Note that $\{\hat{A}, \hat{B}\}$ denotes the anti-commutator of two operators $\hat{A}$ and $\hat{B}$ defined as $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$

Problem 1.7

Prove the following :

(a) $[\hat{L}_z, \cos \phi] = i\hbar \sin \phi$

(b) $[\hat{L}_z, \sin 2\phi] = 2i\hbar(-\cos^2 \phi + \sin^2 \phi)$

(c) $\hat{L}_z(\cos^2 \phi - \sin^2 \phi + 2i \sin \phi \cos \phi) = 2\hbar e^{2i\phi}$

(d) $[\hat{L}_-, e^{-i\phi} \sin \theta] = 0$

(e) $[\hat{L}_-, \cos \theta] = \hbar e^{-i\phi} \sin \theta$

Hint : Use the form of the operators in spherical polar coordinates. Another way would be to change everything in Cartesian coordinates and do the same.

1.4 General Formalism of Angular Momentum

In this section we will do a most generic formalism of angular momentum. To do this we have go through a bit of algebra. But the “essence” of this section will be very useful for the future discussions and also for the current understanding of it.

Consider a general angular momentum operator $\hat{\mathbf{J}}$ defined by it's three components $\hat{J}_x$, $\hat{J}_y$ and $\hat{J}_z$. They satisfy the same commutation algebra as for the case of angular momentum.

$$[\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z, \quad [\hat{J}_y, \hat{J}_z] = i\hbar \hat{J}_x, \quad [\hat{J}_z, \hat{J}_x] = i\hbar \hat{J}_y \quad (1.24)$$

Which can also be written in a compact notation :

$$[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k \quad (1.25)$$

From the commutation relation we conclude that they do not have common eigenstates as they do not commute with each other. The square of angular momentum is defined as,

$$\hat{\mathbf{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 \quad (1.26)$$

It commutes with $\hat{J}_x$, $\hat{J}_y$, $\hat{J}_z$, or equivalently $[\hat{\mathbf{J}}^2, \hat{J}_k] = 0$. One thing to consider that as they represent some “physical observables” they all should be hermitian in order to have real eigenvalues.

Now coming to the point. What we wanna do ? We want to find the eigenvalue and eigenstates corresponding to those operators. Now as $\hat{\mathbf{J}}^2$ commutes with all three components ($\hat{J}_x$, $\hat{J}_y$, $\hat{J}_z$) they can be separately diagonalized with $\hat{\mathbf{J}}^2$. But as they do not commute themselves, we can choose only one of them to be simultaneously diagonalized with $\hat{\mathbf{J}}^2$. For convenience we take $\hat{J}_z$ here. Now one thing you might think “what's so special about ‘z’ axis”. The answer is “it isn't”. We could have proceed to find the common eigenstates for $\hat{J}_x$ and $\hat{\mathbf{J}}^2$ or $\hat{J}_y$ and $\hat{\mathbf{J}}^2$.

Problem 1.8 Jacobi Identity

Prove the following :

- (a) $[\hat{J}_x, [\hat{J}_y, \hat{J}_z]] + [\hat{J}_y, [\hat{J}_z, \hat{J}_x]] + [\hat{J}_z, [\hat{J}_x, \hat{J}_y]] = 0$
- (b) $[\hat{J}_x^2, \hat{J}_y^2] = [\hat{J}_y^2, \hat{J}_z^2] = [\hat{J}_z^2, \hat{J}_x^2]$

1.4.1 Summary of what we wanna do

I hope you have understood about what we are doing here. But for once let me summarize. “We want to find the common eigenstates of operators $\hat{\mathbf{J}}^2$ and $\hat{J}_z$, and also the corresponding eigenvalues.” In the following few pages we have to go through some “nasty” algebra. But most importantly this will give us truly some “remarkable results”.

Example 1.2

Proof the following :

$$\hat{\mathbf{J}} \times \hat{\mathbf{J}} = i\hbar \hat{\mathbf{J}} \quad (1.27)$$

Ans:

$$\begin{aligned} \text{L.H.S.} = \hat{\mathbf{J}} \times \hat{\mathbf{J}} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \hat{J}_x & \hat{J}_y & \hat{J}_z \\ \hat{J}_x & \hat{J}_y & \hat{J}_z \end{vmatrix} \\ &= \hat{\mathbf{i}} (\hat{J}_y \hat{J}_z - \hat{J}_z \hat{J}_y) + \hat{\mathbf{j}} (\hat{J}_z \hat{J}_x - \hat{J}_x \hat{J}_z) + \hat{\mathbf{k}} (\hat{J}_x \hat{J}_y - \hat{J}_y \hat{J}_x) \\ &= \hat{\mathbf{i}} [\hat{J}_y, \hat{J}_z] + \hat{\mathbf{j}} [\hat{J}_z, \hat{J}_x] + \hat{\mathbf{k}} [\hat{J}_x, \hat{J}_y] \\ &= i\hbar (\hat{J}_x \hat{\mathbf{i}} + \hat{J}_y \hat{\mathbf{j}} + \hat{J}_z \hat{\mathbf{k}}) \\ &= i\hbar \hat{\mathbf{J}} = \text{R.H.S. (QED)} \end{aligned}$$

1.5 Going through the algebra

All right, let's get back to business. We were looking for joint eigenstates⁶ of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ and corresponding eigenvalues. Denoting joint eigenstates by $|\lambda, \mu\rangle$ where the eigenvalues of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ are given by $\lambda \hbar^2$ and $\mu \hbar$ respectively. i.e.

$$\hat{\mathbf{J}}^2 |\lambda, \mu\rangle = \lambda \hbar^2 |\lambda, \mu\rangle \quad (1.28a)$$

$$\hat{J}_z |\lambda, \mu\rangle = \mu \hbar |\lambda, \mu\rangle \quad (1.28b)$$

Notice that the factor $\hbar$ is introduced so that λ and μ becomes dimensionless, i.e. purely a number. Also consider that states are orthonormal, i.e.

$$\langle \lambda', \mu' | \lambda, \mu \rangle = \delta_{\lambda, \lambda'} \delta_{\mu, \mu'} \quad (1.29)$$

1.5.1 Ladder Operator Technique

In order to find the eigenvalues λ and μ , we will use “Ladder Operator” technique. We define the ladder operators by :

$$\hat{J}_{\pm} = \hat{J}_x \pm i\hat{J}_y \quad (1.30)$$

⁶Same as the term “common eigenstates”.

We see that by Eq. (1.30) we have :

$$\hat{J}_x = \frac{1}{2} (\hat{J}_+ + \hat{J}_-) \quad (1.31a)$$

$$\hat{J}_y = \frac{1}{2i} (\hat{J}_+ - \hat{J}_-) \quad (1.31b)$$

Problem 1.9 “Hermitianity” of $\hat{J}_\pm$

Are $\hat{J}_\pm$ hermitian ?? If not find their hermitian conjugates.

Hint: Use the definition as per Eq. (1.30) and take a dagger ($\dagger$) on both sides.

Problem 1.10

Prove the following commutation relations :

(a) $[\hat{\mathbf{J}}^2, \hat{J}_\pm] = 0$

(b) $[\hat{J}_+, \hat{J}_-] = 2\hbar\hat{J}_z$

Problem 1.11

Using Eq. (1.31) write down the operators $\hat{J}_x^2$ and $\hat{J}_y^2$ in terms of ladder operators $\hat{J}_\pm$.

Example 1.3

Prove that :

$$[\hat{J}_z, \hat{J}_\pm] = \pm\hbar\hat{J}_\pm \quad (1.32)$$

Proof:

$$\begin{aligned} \text{L.H.S.} &= [\hat{J}_z, \hat{J}_\pm] \\ &= [\hat{J}_z, \hat{J}_x \pm i\hat{J}_y] \\ &= [\hat{J}_z, \hat{J}_x] \pm i[\hat{J}_z, \hat{J}_y] \\ &= i\hbar\hat{J}_y \pm i(-i\hbar\hat{J}_x) \\ &= \mp i^2\hbar\hat{J}_x + i\hbar\hat{J}_y \\ &= \pm\hbar\hat{J}_x + i\hbar\hat{J}_y \\ &= \pm\hbar(\hat{J}_x \pm i\hat{J}_y) \\ &= \pm\hbar\hat{J}_\pm = \text{R.H.S. (QED)} \end{aligned}$$

1.5.2 Action of $\hat{J}_\pm$

Consider a state which is obtained by the action of $\hat{J}_\pm$ on $|\lambda, \mu\rangle$, i.e. $\hat{J}_\pm |\lambda, \mu\rangle$. Now our question is “Is this state we have obtained an eigenstate of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ ” and however if so then “What is the corresponding eigenvalues”. In order to do

answer these question we proceed in following way :

$$\begin{aligned}
 \hat{\mathbf{J}}^2 \left[\hat{J}_{\pm} |\lambda, \mu\rangle \right] &= \hat{\mathbf{J}}^2 \hat{J}_{\pm} |\lambda, \mu\rangle \\
 &= \hat{J}_{\pm} \hat{\mathbf{J}}^2 |\lambda, \mu\rangle^7 \\
 &= \hat{J}_{\pm} \left[\hat{\mathbf{J}}^2 |\lambda, \mu\rangle \right] \\
 &= \hat{J}_{\pm} \left[\lambda \hbar^2 |\lambda, \mu\rangle \right] \\
 &= \lambda \hbar^2 \left[\hat{J}_{\pm} |\lambda, \mu\rangle \right]
 \end{aligned}$$

So therefore we conclude that if $|\lambda, \mu\rangle$ is an eigenstate of $\hat{\mathbf{J}}^2$ with an eigenvalue $\lambda \hbar^2$ then, so does $\hat{J}_{\pm} |\lambda, \mu\rangle$ having the same eigenvalue. Now as $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ commute we are ascertain that $\hat{J}_{\pm} |\lambda, \mu\rangle$ will also be an eigenstate of the operator $\hat{J}_z$. But what is the “corresponding eigenvalue”. Let us go through it explicitly. We proceed the very similar way as we did before :

$$\begin{aligned}
 \hat{J}_z \left[\hat{J}_{\pm} |\lambda, \mu\rangle \right] &= \hat{J}_z \hat{J}_{\pm} |\lambda, \mu\rangle \\
 &= \hat{J}_{\pm} \hat{J}_z |\lambda, \mu\rangle \pm \hbar \hat{J}_{\pm} |\lambda, \mu\rangle^8 \\
 &= \hat{J}_{\pm} \left[\hat{J}_z |\lambda, \mu\rangle \right] \pm \hbar \hat{J}_{\pm} |\lambda, \mu\rangle \\
 &= \hat{J}_{\pm} \left[\mu \hbar |\lambda, \mu\rangle \right] \pm \hbar \hat{J}_{\pm} |\lambda, \mu\rangle \\
 &= (\mu \pm 1) \hbar \left[\hat{J}_{\pm} |\lambda, \mu\rangle \right]
 \end{aligned}$$

So we see that if we act $\hat{J}_{\pm}$ on the state $|\lambda, \mu\rangle$, the magnitude square of vector $\lambda \hbar^2$ remains the same but the ‘z’ component of angular momentum is changed by $\pm \hbar$ unit. That is why we call

$$\begin{aligned}
 \hat{J}_+ &\rightarrow \text{Raising Operator} \\
 \hat{J}_- &\rightarrow \text{Lowering Operator}
 \end{aligned}$$

as $\hat{J}_+$ increases the eigenvalue of $\hat{J}_z$ by one $\hbar$ unit and $\hat{J}_-$ decreases the eigenvalue of $\hat{J}_z$ by the same value. Note that both states having the same eigenvalue for the $\hat{\mathbf{J}}^2$ operator i.e. $\lambda \hbar^2$. So for a given value of ‘ λ ’, we get ladder of states with each rung separated from its’ neighbours by $\hbar$ unit in the eigenvalue of $\hat{J}_z$ ⁹. To ascend the ladder we need to apply $\hat{J}_+$ and to descend $\hat{J}_-$. But this can’t go on forever. Eventually we will reach a state for which eigenvalue of $\hat{J}_z$ ¹⁰ will exceed the total value. So we must have the expectation value of the operator $\hat{\mathbf{J}}^2 - \hat{J}_z^2$ to be positive for all the eigenstates. In other terms :

$$\langle \hat{\mathbf{J}}^2 \rangle = \langle \hat{J}_x^2 \rangle + \langle \hat{J}_y^2 \rangle + \langle \hat{J}_z^2 \rangle \Rightarrow \lambda \hbar^2 \geq \mu^2 \hbar^2 \text{ or } \lambda \geq \mu^2^{11} \quad (1.33)$$

Now one thing is that the eigenvalue of operator $\hat{\mathbf{J}}^2$ must be positive. i.e. $\lambda \geq 0$. Now the inequality as given in Eq. (1.33) must hold for the allowed $|\lambda, \mu\rangle$ states. If that so eigenvalue of $\hat{J}_z$ has a upper limit say $\mu_{\max}$ upto which this holds for a given λ . Beyond which the inequality will no longer hold and if we apply the raising operator on this state, this should not produce any new state, as it will be violating the inequality. So based on these arguments we say that “there should not exist any state” which will have eigenvalue of $\hat{J}_z$ as $(\mu_{\max} + 1)\hbar$. In simple terms if $|\lambda, \mu_{\max}\rangle$ is the above

⁷Here we have used the fact that $[\hat{\mathbf{J}}^2, \hat{J}_{\pm}] = 0$ which gives $\hat{\mathbf{J}}^2 \hat{J}_{\pm} = \hat{J}_{\pm} \hat{\mathbf{J}}^2$

⁸Here we have used the fact that $[\hat{J}_z, \hat{J}_{\pm}] = \pm \hbar \hat{J}_{\pm}$ which gives $\hat{J}_z \hat{J}_{\pm} = \hat{J}_{\pm} \hat{J}_z \pm \hbar \hat{J}_{\pm}$

⁹So in “quantum mechanics” the spectrum is discrete, with a difference of $\hbar$.

¹⁰More precisely we mean absolute value of it.

¹¹This is as $\langle \hat{J}_x^2 \rangle = \langle \lambda, \mu | \hat{J}_x^2 | \lambda, \mu \rangle \geq 0$ and same goes for $\langle \hat{J}_y^2 \rangle$. Here $\langle \hat{J}_z^2 \rangle = \langle \lambda, \mu | \hat{J}_z^2 | \lambda, \mu \rangle = \mu^2 \hbar^2$.

mentioned state then “it cannot be raised further”. In mathematical terms we have :

$$\hat{J}_+ |\lambda, \mu_{\max}\rangle = 0 \quad (1.34a)$$

$$\hat{\mathbf{J}}^2 |\lambda, \mu_{\max}\rangle = \lambda \hbar^2 |\lambda, \mu_{\max}\rangle \quad (1.34b)$$

$$\hat{J}_z |\lambda, \mu_{\max}\rangle = \mu_{\max} \hbar |\lambda, \mu_{\max}\rangle \quad (1.34c)$$

Example 1.4

Find out the value of $\hat{J}_\pm \hat{J}_\mp$ and write $\hat{\mathbf{J}}^2$ in terms of this.

$$\begin{aligned} \hat{J}_\pm \hat{J}_\mp &= (\hat{J}_x \pm i\hat{J}_y)(\hat{J}_x \mp i\hat{J}_y) \\ &= \hat{J}_x^2 + \hat{J}_y^2 \pm i\hat{J}_y\hat{J}_x \mp i\hat{J}_y\hat{J}_x \\ &= \hat{J}_x^2 + \hat{J}_y^2 \pm i(\hat{J}_y\hat{J}_x - \hat{J}_y\hat{J}_x) \\ &= \hat{\mathbf{J}}^2 - \hat{J}_z^2 \mp i[\hat{J}_x, \hat{J}_y] \\ &= \hat{\mathbf{J}}^2 - \hat{J}_z^2 \mp i(i\hbar\hat{J}_z) \\ &= \hat{\mathbf{J}}^2 - \hat{J}_z^2 \pm \hbar\hat{J}_z \end{aligned} \quad (1.35)$$

Which gives :

$$\hat{\mathbf{J}}^2 = \hat{J}_\pm \hat{J}_\mp + \hat{J}_z^2 \mp \hbar\hat{J}_z \quad (1.36)$$

Now consider the following :

$$\begin{aligned} \hat{\mathbf{J}}^2 |\lambda, \mu_{\max}\rangle &= (\hat{J}_- \hat{J}_+ + \hat{J}_z^2 + \hbar\hat{J}_z) |\lambda, \mu_{\max}\rangle \\ &= \hat{J}_- \underbrace{[\hat{J}_+ |\lambda, \mu_{\max}\rangle]}_0 + \hat{J}_z^2 |\lambda, \mu_{\max}\rangle + \hbar\hat{J}_z |\lambda, \mu_{\max}\rangle \\ &= \mu_{\max}^2 \hbar^2 |\lambda, \mu_{\max}\rangle + \mu_{\max} \hbar^2 |\lambda, \mu_{\max}\rangle \\ &= \mu_{\max}(\mu_{\max} + 1) \hbar^2 |\lambda, \mu_{\max}\rangle \end{aligned}$$

Comparing the last equation and Eq. (1.34b) we get the following relation :

$$\lambda = \mu_{\max}(\mu_{\max} + 1) \quad (1.37)$$

Eq. (1.37) tells us the the eigenvalue of $\hat{\mathbf{J}}^2$ in terms of the “maximum eigenvalue” of $\hat{J}_z$. In a very similar way if we keep acting $\hat{J}_-$ on the state $|\lambda, \mu\rangle$ and by doing this repeatedly we decrease the eigenvalue of $\hat{J}_z$ by $\hbar$ unit per action. We will eventually hit the “bottom rung” of the ladder say $\hat{J}_z$ eigenvalue is $\mu_{\min} \hbar$ and we denote the state by $|\lambda, \mu_{\min}\rangle$. By the same token :

$$\hat{J}_- |\lambda, \mu_{\min}\rangle = 0 \quad (1.38a)$$

$$\hat{\mathbf{J}}^2 |\lambda, \mu_{\min}\rangle = \lambda \hbar^2 |\lambda, \mu_{\min}\rangle \quad (1.38b)$$

$$\hat{J}_z |\lambda, \mu_{\min}\rangle = \mu_{\min} \hbar |\lambda, \mu_{\min}\rangle \quad (1.38c)$$

The very similar way as we did before :

$$\begin{aligned}
 \hat{\mathbf{J}}^2 |\lambda, \mu_{\min}\rangle &= (\hat{J}_+ \hat{J}_- + \hat{J}_z^2 - \hbar \hat{J}_z) |\lambda, \mu_{\min}\rangle \\
 &= \hat{J}_+ \underbrace{[\hat{J}_- |\lambda, \mu_{\min}\rangle]}_0 + \hat{J}_z^2 |\lambda, \mu_{\min}\rangle - \hbar \hat{J}_z |\lambda, \mu_{\min}\rangle \\
 &= \mu_{\min}^2 \hbar^2 |\lambda, \mu_{\min}\rangle - \mu_{\min} \hbar^2 |\lambda, \mu_{\min}\rangle \\
 &= \mu_{\min}(\mu_{\min} - 1) \hbar^2 |\lambda, \mu_{\min}\rangle
 \end{aligned}$$

which gives :

$$\lambda = \mu_{\min}(\mu_{\min} - 1) \quad (1.39)$$

Comparing Eq. (1.37) and Eq. (1.39) we get :

$$\begin{aligned}
 \mu_{\max}(\mu_{\max} + 1) &= \mu_{\min}(\mu_{\min} - 1) \\
 \Rightarrow \begin{cases} \mu_{\min} = -\mu_{\max} \\ \mu_{\min} = \mu_{\max} + 1 \end{cases} & \quad (1.40)
 \end{aligned}$$

Out of these two cases $\mu_{\min} = \mu_{\max} + 1$ cannot be true as “bottom rung will be higher than the top rung”. Finally we have :

$$\mu_{\min} = -\mu_{\max} \quad (1.41)$$

For now, make a mental picture like this ; given a state $|\lambda, \mu\rangle$, while acted upon by $\hat{J}_+$, it increases $\hat{J}_z$ eigenvalue by $\pm\hbar$ unit. But it can't go on forever. We will eventually hit the top (bottom) rung by the repeated application of $\hat{J}_+$ ($\hat{J}_-$). All we know for now is that $\mu_{\min} = -\mu_{\max}$. But what about the nature of this $\mu_{\max}$ or the $\mu_{\min}$. All we know for sure they are “real”. But are they “integer or fractional ???”. We just don't know it yet.

Now in order to find the answer to this question, think this “given the state $|\lambda, \mu_{\max}\rangle$, how are we gonna get the state $|\lambda, \mu_{\min}\rangle$ ”. The answer is very simple; By successive application of $\hat{J}_-$ right !!! Each time it will descend the ladder by $\hbar$ unit, if we talk about the eigenvalue of $\hat{J}_z$. Now suppose, in order to get from top to bottom we had to apply $\hat{J}_-$ operator N times, where N is a non negative integer. So by N times it will decrease the eigenvalue of $\hat{J}_z$ by $N\hbar$ amount. So we have :

$$\mu_{\min} \hbar = \mu_{\max} \hbar - N \hbar$$

Using Eq. (1.41) we get :

$$\mu_{\max} = -\mu_{\min} = \frac{N}{2} \quad (1.42)$$

Calling $\mu_{\max} = j$, we get $j = N/2$. By this we came to the conclusion that “ $\mu_{\max}$ can be integer or half-integer depending on the value of N being either odd or even”. Now on the state $|\lambda, \mu_{\max}\rangle$ if we apply $\hat{J}_-$ operator M times (surely $M \leq N$), then we have the state with $\hat{J}_z$ eigenvalue as $\mu_M = \mu_{\max} - M$ in units of $\hbar$. So the eigenvalue we talked about previously, μ is also either “integer” or “half-integer”. And the eigenvalue of $\hat{\mathbf{J}}^2$ is given by $\lambda = j(j+1)$ in units of $\hbar^2$. Writing μ as m , we denote the state $|\lambda, \mu\rangle$ by $|j, m\rangle$, where the allowed values of “ m ” are $-j \leq m \leq j$ with an increment in steps of “1” and also $j \geq 0$. Finally we can write Eq. (1.28) as :

$$\hat{\mathbf{J}}^2 |j, m\rangle = j(j+1) \hbar^2 |j, m\rangle \quad (1.43a)$$

$$\hat{J}_z |j, m\rangle = m \hbar |j, m\rangle \quad (1.43b)$$

Where $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. And for a given j the allowed values of m are :

$$m = -j, -j+1, \dots, j-1, j \quad (1.44)$$

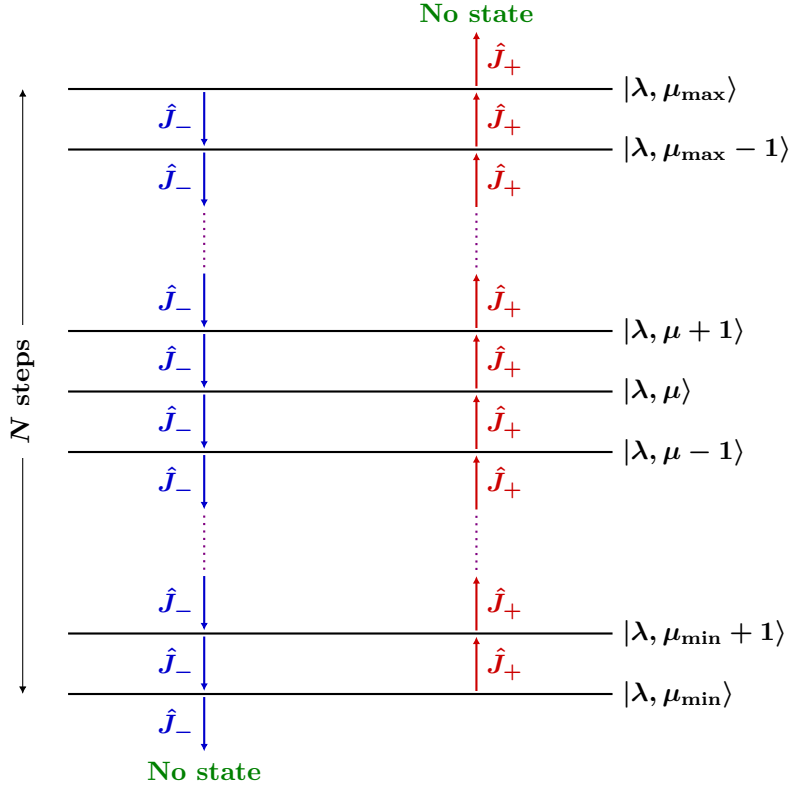


Figure 1.1: The “ladder”.

So there are $(2j + 1)$ values of m for a given j . For example if $j = 1$ then we have $m = +1, 0, -1$. So this gives “three states”, $|1, +1\rangle$, $|1, 0\rangle$ and $|1, -1\rangle$. i.e. $2j + 1$ no of “rungs” on the ladder. The spectra of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ are discrete. The orthonormality condition is given by Eq. (1.53a). This is schematically shown in Fig (1.1).

1.5.3 Finding out the coefficients

From the previous discussions it is very clear that :

$$\hat{J}_{\pm} |j, m\rangle = C_{jm}^{\pm} |j, m \pm 1\rangle \quad (1.45)$$

Our job is to find out the coefficient $C_{jm}^{\pm}$ so that given the normalized state $|j, m\rangle$ we end up with the normalized state $|j, m \pm 1\rangle$. Taking $\dagger$ on both sides of Eq. (1.45) we obtain :

$$\langle j, m | \hat{J}_{\mp} = (C_{jm}^{\pm})^* \langle j, m \pm 1 | \quad (1.46)$$

Using the normalization condition and by Eqs. (1.45), (1.46) and (1.35) we get :

$$\begin{aligned} |C_{jm}^{\pm}|^2 &= \langle j, m | \hat{J}_{\mp} \hat{J}_{\pm} |j, m\rangle \\ &= \langle j, m | \hat{\mathbf{J}}^2 |j, m\rangle - \langle j, m | \hat{J}_z^2 |j, m\rangle \mp \hbar \langle j, m | \hat{J}_z |j, m\rangle \\ &= j(j+1)\hbar^2 - m^2\hbar^2 \mp m\hbar^2 \\ &= \hbar^2 [j(j+1) - m^2 \mp m] \\ &= \hbar^2 [j(j+1) - m(m \pm 1)] \\ &= \hbar^2 [(j \mp m)(j \pm m + 1)] \end{aligned}$$

Assuming the coefficients to be real we finally obtain :

$$C_{jm}^{\pm} = \hbar \sqrt{j(j+1) - m(m \pm 1)} = \hbar \sqrt{(j \mp m)(j \pm m + 1)} \quad (1.47)$$

Rewriting Eq. (1.45) we get :

$$\hat{J}_{\pm} |j, m\rangle = \begin{cases} \hbar \sqrt{j(j+1) - m(m \pm 1)} & |j, m \pm 1\rangle \\ \hbar \sqrt{(j \mp m)(j \pm m + 1)} & |j, m \pm 1\rangle \end{cases} \quad (1.48)$$

Problem 1.12 Action of $\hat{J}_x$ and $\hat{J}_y$ on $|j, m\rangle$

Using the definitions of $\hat{J}_x$ and $\hat{J}_y$ given in Eq. (1.31) and Eq. (1.48) find out :

(a) $\hat{J}_x |j, m\rangle$

(b) $\hat{J}_y |j, m\rangle$

Problem 1.13 Expectation value of $\hat{J}_x$ and $\hat{J}_y$

Find out the expectation value of $\hat{J}_x$ and $\hat{J}_y$, i.e. $\langle \hat{J}_x \rangle$ and $\langle \hat{J}_y \rangle$.

Hint : Use the results from **Problem (1.12)**. They both are zero.

Example 1.5 Expectation value of $\hat{J}_x^2$ and $\hat{J}_y^2$

Find out the expectation value of $\hat{J}_x^2$ and $\hat{J}_y^2$, i.e. $\langle \hat{J}_x^2 \rangle$ and $\langle \hat{J}_y^2 \rangle$. Using Eq. (1.31) we get :

$$\begin{aligned} \hat{J}_x^2 &= +\frac{1}{4} [\hat{J}_+^2 + \hat{J}_-^2 + \hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+] \\ \hat{J}_y^2 &= -\frac{1}{4} [\hat{J}_+^2 + \hat{J}_-^2 - \hat{J}_+ \hat{J}_- - \hat{J}_- \hat{J}_+] \end{aligned}$$

Way 1 :

As we know $\langle \hat{J}_+^2 \rangle = \langle j, m | \hat{J}_+^2 | j, m \rangle = 0$ and very similarly we have $\langle \hat{J}_-^2 \rangle = \langle j, m | \hat{J}_-^2 | j, m \rangle = 0$, we conclude that $\langle \hat{J}_x^2 \rangle = \langle \hat{J}_y^2 \rangle$. Using Eqs. (1.28) and (1.33) we therefore get :

$$\langle \hat{J}_x^2 \rangle = \langle \hat{J}_y^2 \rangle = \frac{1}{2} [\langle \hat{\mathbf{J}}^2 \rangle - \langle \hat{J}_z^2 \rangle] = \frac{\hbar^2}{2} [j(j+1) - m^2] \quad (1.49)$$

Way 2:

Again using Eq. (1.48) and the last equation we first compute :

$$\begin{aligned} \hat{J}_{\pm} \hat{J}_{\mp} |j, m\rangle &= \hat{J}_{\pm} \left[\hbar \sqrt{j(j+1) - m(m \mp 1)} |j, m \mp 1\rangle \right] \\ &= \hbar \sqrt{j(j+1) - m(m \mp 1)} \left[\hat{J}_{\pm} |j, m \mp 1\rangle \right] \\ &= \hbar \sqrt{j(j+1) - m(m \mp 1)} \times \hbar \sqrt{j(j+1) - (m \mp 1)(m \mp 1 \pm 1)} |j, m\rangle \\ &= \hbar^2 (j(j+1) - m(m \mp 1)) |j, m\rangle \end{aligned}$$

Which gives :

$$\langle \hat{J}_{\pm} \hat{J}_{\mp} \rangle = \hbar^2 (j(j+1) - m(m \mp 1)) \quad (1.50)$$

¹²If we put $m = j$ ($-j$) for the $\hat{J}_+$ ($\hat{J}_-$) you get the coefficient to be zero. That's expected.

Using this we finally get :

$$\begin{aligned}\langle \hat{J}_x^2 \rangle &= \langle \hat{J}_y^2 \rangle = \frac{1}{4} \left[\langle \hat{J}_+ \hat{J}_- \rangle + \langle \hat{J}_- \hat{J}_+ \rangle \right] \\ &= \frac{\hbar^2}{4} \left[(j(j+1) - m(m-1)) + (j(j+1) - m(m+1)) \right] \\ &= \frac{\hbar^2}{2} \left[j(j+1) - m^2 \right]\end{aligned}$$

Problem 1.14 “Uncertainty Principle”

Using the results obtained in **Example (1.5)** and **Problem (1.13)**

1. Find out ΔJ_x and ΔJ_y . Where $\Delta J_i = \sqrt{\langle \hat{J}_i^2 \rangle - \langle \hat{J}_i \rangle^2}$.
2. Is this consistent with “generalized uncertainty principle”, i.e.

$$\Delta J_x \Delta J_y \geq \frac{\hbar}{2} |\langle \hat{J}_z \rangle| \quad (1.51)$$

Hint:

Using $\Delta J_x \Delta J_y = \frac{\hbar^2}{2} [j(j+1) - m^2]$ and $\langle \hat{J}_z \rangle = m\hbar$ we can write $\frac{\hbar}{2} |\langle \hat{J}_z \rangle| = \frac{|m|\hbar^2}{2}$. Now as $j \geq |m|$ we must have : $\Delta J_x \Delta J_y \geq \frac{\hbar^2}{2} [|m|(|m|+1) - |m|^2] \geq \frac{|m|\hbar^2}{2}$.

1.6 Matrix Representation of Angular Momentum

In the last section Sec. 1.4, the formalism we have developed is independent of any particular representation. So far we have talked about the eigenvalues of $\hat{\mathbf{J}}^2$ and $\hat{J}_z$. There are many ways to represent the angular momentum operators and their eigenstates. In this section we are going to discuss about the matrix representation of them, where “eigenkets” are represented by “column vectors” and “operators” by “square matrices”. Remember that in Sec. 1.3 we had represented “orbital angular momentum” in the position basis in form of “differential operators”.

Denoting the joint eigenstates by $|j, m\rangle$, we can represent the operators $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ in this basis. Note that the basis is “discrete, complete and orthonormal”. For example, $j = \frac{1}{2}$ will give $m = \pm\frac{1}{2}$. So we have two basis states $|\frac{1}{2}, +\frac{1}{2}\rangle$ and $|\frac{1}{2}, -\frac{1}{2}\rangle$ and we can express operators in this basis.

“Orthonormality and completeness” conditions are given by :

$$\langle j', m' | j, m \rangle = \delta_{j,j'} \delta_{m,m'} \quad (1.52a)$$

$$\sum_{m=-j}^j |j, m\rangle \langle j, m| = \hat{I} \quad (1.52b)$$

$\hat{I}$ here is a $(2j+1) \times (2j+1)$ identity matrix.

Now for any operator $\hat{A}$, the matrix element of the representation of the operator in $|j, m\rangle$ basis is given by $A_{m'm} = \langle j, m' | \hat{A} | j, m \rangle$. The operators $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ are diagonal in $|j, m\rangle$ basis as :

$$\langle j, m' | \hat{\mathbf{J}}^2 | j, m \rangle = j(j+1) \hbar^2 \delta_{m,m'} \quad (1.53a)$$

$$\langle j, m' | \hat{J}_z | j, m \rangle = m\hbar \delta_{m,m'} \quad (1.53b)$$

The diagonal elements are being $j(j+1)\hbar^2$ for $\hat{\mathbf{J}}^2$ operator and for $\hat{J}_z$ it is $m\hbar$. The next thing comes naturally in our mind is that “what is the representation of $\hat{J}_\pm$ and $\hat{J}_x, \hat{J}_y$ ”.

1.6.1 Matrix representation of $\hat{J}_\pm, \hat{J}_x$ and $\hat{J}_y$

From the discussions of previous section we got :

$$\hat{J}_\pm |j, m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)} |j, m\pm 1\rangle \quad (1.54)$$

Which gives :

$$\langle j, m' | \hat{J}_\pm | j, m \rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)} \delta_{m', m\pm 1} \quad (1.55)$$

From Eq. (1.55) it is very clear that in $|j, m\rangle$ basis, $\hat{J}_\pm$ is not diagonal. This is because as they do not commute with $\hat{J}_z$ and henceforth can not be simultaneously diagonalized. From Eq. (1.31) we can write the representations of $\hat{J}_x$ and $\hat{J}_y$ given that we know the representation of $\hat{J}_\pm$, Which we will do next by Eq. (1.54).

$$\begin{aligned} \langle j, m' | \hat{J}_x | j, m \rangle &= \frac{1}{2} \left[\langle j, m' | \hat{J}_+ | j, m \rangle + \langle j, m' | \hat{J}_- | j, m \rangle \right] \\ &= \frac{\hbar}{2} \left[\sqrt{j(j+1) - m(m+1)} \delta_{m', m+1} + \sqrt{j(j+1) - m(m-1)} \delta_{m', m-1} \right] \end{aligned} \quad (1.56)$$

And similarly for $\hat{J}_y$,

$$\begin{aligned} \langle j, m' | \hat{J}_y | j, m \rangle &= \frac{1}{2i} \left[\langle j, m' | \hat{J}_+ | j, m \rangle - \langle j, m' | \hat{J}_- | j, m \rangle \right] \\ &= \frac{\hbar}{2i} \left[\sqrt{j(j+1) - m(m+1)} \delta_{m', m+1} - \sqrt{j(j+1) - m(m-1)} \delta_{m', m-1} \right] \end{aligned} \quad (1.57)$$

From Eqs. (1.56) and (1.57) we can see that they are not diagonal. The same statement applies here also, as $\hat{J}_x, \hat{J}_y$ do not commute with $\hat{J}_z$. Now for a given “ j ”, once we find the matrix representation of $\hat{J}_z$, we can then proceed to get the eigenstates for different values of m using the eigenvalue equation

$$\hat{J}_z |j, m\rangle = m\hbar |j, m\rangle \quad (1.58)$$

1.6.2 Structure of matrix for operator $\hat{A}$

$$\hat{A} = \begin{pmatrix} & |j, j\rangle & |j, j-1\rangle & \cdots & |j, -j+1\rangle & |j, -j\rangle \\ \hline \langle j, j| & A_{j,j} & A_{j,j-1} & \cdots & A_{j,-j+1} & A_{j,-j} \\ \langle j, j-1| & A_{j-1,j} & A_{j-1,j-1} & \cdots & A_{j-1,-j+1} & A_{j-1,-j} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \langle j, -j+1| & A_{-j+1,j} & A_{-j+1,j-1} & \cdots & A_{-j+1,-j+1} & A_{-j+1,-j} \\ \langle j, -j| & A_{-j,j} & A_{-j,j-1} & \cdots & A_{-j,-j+1} & A_{-j,-j} \end{pmatrix} \quad (1.59)$$

Where we have defined :

$$A_{m,m'} = \langle j, m | \hat{A} | j, m' \rangle, \quad m, m' = j, j-1, \dots, -j \quad (1.60)$$

This structure is followed from now on, that we will not show explicitly henceforth.

1.6.3 Steps to compute the representations for a given j

- (a) First write $\hat{\mathbf{J}}^2 = j(j+1)\hbar^2 \hat{I}$

(b) Then write $\hat{J}_z = \hbar \begin{pmatrix} j & 0 & 0 & 0 & 0 \\ 0 & j-1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & -j+1 & 0 \\ 0 & 0 & 0 & 0 & -j \end{pmatrix}$

(c) Use Eq. (1.55) to get the representation of $\hat{J}_{\pm}$.

(d) You can either use Eqs. (1.56) and (1.57) to get the representation for $\hat{J}_x, \hat{J}_y$ or using $\hat{J}_{\pm}$ representation done in (c) and using Eq. (1.31).

Problem 1.15

Consider the case where $j = 1$.

- Find the matrices representing $\hat{\mathbf{J}}^2, \hat{J}_z, \hat{J}_{\pm}, \hat{J}_x, \hat{J}_y$.
- Using the matrices $\hat{J}_x, \hat{J}_y, \hat{J}_z$ calculate the commutations $[\hat{J}_x, \hat{J}_y], [\hat{J}_y, \hat{J}_z], [\hat{J}_z, \hat{J}_x]$.
- Verify $(\hat{J}_{\pm})^{\dagger} = \hat{J}_{\mp}$ ¹³
- Find out the eigenvalues and eigenvectors corresponding to $\hat{J}_x$ and $\hat{J}_y$.

Example 1.6

Find out joint eigenstates of $\hat{J}_z$ and $\hat{\mathbf{J}}^2$ for $j = 1$. Also verify that they indeed form a complete orthonormal basis.

Ans: For $j = 1$, we have allowed values of m are 1, 0, -1. We denote the joint eigenstates by $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$. The matrix rep of $\hat{\mathbf{J}}^2, \hat{J}_z$ are given :

$$\hat{\mathbf{J}} = 1(1+1)\hbar^2 \hat{I}_{3 \times 3} = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (1.61)$$

$$\hat{J}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (1.62)$$

Writing $|j, m\rangle = \begin{pmatrix} a_m \\ b_m \\ c_m \end{pmatrix}$ and using Eq. (1.58) we get :

$$\hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_m \\ b_m \\ c_m \end{pmatrix} = m\hbar \begin{pmatrix} a_m \\ b_m \\ c_m \end{pmatrix}$$

$$\Rightarrow \begin{cases} a_m = ma_m \\ 0 = mb_m \\ c_m = mc_m \end{cases} \quad (1.63)$$

Now for $m = 1$ we got $a_1 = 1, b_1 = 0, c_1 = 0$ and similarly $m = -1$ we got $a_{-1} = 0, b_{-1} = 0, c_{-1} = 1$. For

¹³For a matrix A , $A^{\dagger} = (A^T)^*$

$m = 0$ gives $a_0 = 0$, $c_0 = 0$ and b_0 is indeterminate, for normalized state we take $b_0 = 1$. So we finally have :

$$|1, 1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1, 0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1, -1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}^{14} \quad (1.64)$$

We can easily see that they are orthonormal, i.e.

$$\langle 1, m' | 1, m \rangle = \delta_{m, m'}$$

And also for the completeness :

$$\begin{aligned} \sum_{m=-1}^{m=1} |1, m\rangle\langle 1, m| &= |1, 1\rangle\langle 1, 1| + |1, 0\rangle\langle 1, 0| + |1, -1\rangle\langle 1, -1| \\ &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \hat{I}_{3 \times 3} \end{aligned}$$

Problem 1.16

Consider the case $j = 1$. Find out the matrix for the component of $\vec{\mathbf{J}}$ along a unit vector $\hat{\mathbf{n}}$. Find also eigenvalue and eigenvectors.

Hint: Write $\hat{\mathbf{J}} = \hat{J}_x \hat{\mathbf{i}} + \hat{J}_y \hat{\mathbf{j}} + \hat{J}_z \hat{\mathbf{k}}$ and $\hat{\mathbf{n}} = (\sin \theta \cos \phi) \hat{\mathbf{i}} + (\sin \theta \sin \phi) \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}$. Now using the form of $\hat{J}_x$, $\hat{J}_y$ and $\hat{J}_z$ find out $\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}$. The matrix is :

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{J}} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} \sqrt{2} \cos \theta & e^{-i\phi} \sin \theta & 0 \\ e^{i\phi} \sin \theta & 0 & e^{-i\phi} \sin \theta \\ 0 & e^{i\phi} \sin \theta & -\sqrt{2} \cos \theta \end{pmatrix} \quad (1.65)$$

The eigenvalues are $\lambda_1 = \hbar$, $\lambda_2 = 0$, $\lambda_3 = -\hbar$.

Now the eigenvectors are given by :

$$|\lambda_1\rangle = \frac{1}{2} \begin{bmatrix} (1 + \cos \theta)e^{-i\phi} \\ \sqrt{2} \sin \theta \\ (1 - \cos \theta)e^{i\phi} \end{bmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} -\sin \theta e^{-i\phi} \\ \sqrt{2} \cos \theta \\ \sin \theta e^{i\phi} \end{bmatrix}, \quad |\lambda_3\rangle = \frac{1}{2} \begin{bmatrix} (1 - \cos \theta)e^{-i\phi} \\ -\frac{1}{\sqrt{2}} \sin \theta \\ (1 + \cos \theta)e^{i\phi} \end{bmatrix} \quad (1.66)$$

Please verify that they are “complete and orthonormal”. i.e. :

$$\begin{aligned} \langle \lambda_i | \lambda_j \rangle &= \delta_{ij} \\ \sum_{k=1}^3 |\lambda_k\rangle\langle \lambda_k| &= \mathbb{I}_{3 \times 3} \end{aligned}$$

Twist

¹⁴For a given j you can straight away write down representations of $|j, m\rangle$ for different values of m . The only thing you need to use is Eq. (1.58), which gives $|j, j\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $|j, j-1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\dots$, $|j, -j+1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, $|j, -j\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. So the trick is for a given j write the uppermost state $|j, j\rangle$ as a column vector of shape $(2j+1) \times 1$ with first element being 1 and all the other entries are “zero”. Then lower the m value by 1. i.e. $m = j-1$. Write state $|j, j-1\rangle$ with second element being 1 and others are “zero”. Keep doing this you will get all the states. You might think “why is it always so”. The answer lies in Eq. (1.58) and the diagonal structure of $\hat{J}_z$. See **Example (1.6)** to have a better understanding.

If we put $\hat{\mathbf{n}} = \hat{\mathbf{k}}$, then we get $\hat{\mathbf{n}} \cdot \hat{\mathbf{J}} = \hat{J}_z$. Putting $\theta = 0$ in Eq. (1.65) we get the matrix as :

$$\hat{J}_z = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\sqrt{2} \end{pmatrix} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Which is exact form we get in the previous section. Following Eq. (1.66) we write eigenvectors as :

$$|\lambda_1\rangle = \frac{1}{2} \begin{bmatrix} 2e^{-i\phi} \\ 0 \\ 0 \end{bmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ \sqrt{2} \\ 0 \end{bmatrix}, \quad |\lambda_3\rangle = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ 2e^{i\phi} \end{bmatrix}$$

Now these are the exactly the same eigenvectors as we have found out in **Example (1.6)**, but with a “phase factor” $e^{\pm i\phi}$ which is arbitrary anyway.

Now by putting $\hat{\mathbf{n}} = \hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ respectively and find out the form of $\hat{J}_x$ and $\hat{J}_y$. For $\hat{\mathbf{n}} = \hat{\mathbf{i}}$ we have $\phi = 0, \theta = \frac{\pi}{2}$ and for $\hat{\mathbf{n}} = \hat{\mathbf{j}}$ we have $\phi = \frac{\pi}{2}, \theta = \frac{\pi}{2}$. Compare your results with the results you had obtained in **Problem (1.15)**. Also match the “eigenvectors”.

Quick Check

$$\hat{J}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{J}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

Problem 1.17 For generic ‘j’

Work out the matrices $\hat{J}_x$ and $\hat{J}_y$ for a given j .

Hint: Use Eq. (1.55) to construct the matrices $\hat{J}_{\pm}$ and using them construct $\hat{J}_x$ and $\hat{J}_y$. Note that all these matrices have shape $(2j+1) \times (2j+1)$. For instance we can show that the structure of $\hat{J}_{\pm}$ are as follows :

$$\hat{J}_+ = \begin{pmatrix} 0 & b_j & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & b_{j-1} & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & b_{j-2} & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & b_{-j+1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

$$\hat{J}_- = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ b_j & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & b_{j-1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & b_{j-2} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & b_{-j+1} & 0 \end{pmatrix}$$

Where $b_r \equiv \hbar\sqrt{(j+r)(j+1-r)}$.

Problem 1.18 Computing matrices given ‘ j ’

Write a code which takes j as input and gives $\hat{J}_x, \hat{J}_y, \hat{J}_z, \hat{\mathbf{J}}^2$ and $\hat{J}_\pm$ matrices. You can work in “natural units” so that $\hbar = 1$. Please note that j can be either integer or half integer and most importantly can not be negative.

1.7 Geometric Representation of Angular Momentum

For a fixed value of j , the angular momentum vector $\hat{\mathbf{J}}$ may be represented by a vector of length $\sqrt{\langle \hat{\mathbf{J}}^2 \rangle} = \hbar\sqrt{j(j+1)}$, whose “ z ” component is given by $\langle \hat{J}_z \rangle = m\hbar$. Notice that the “magnitude” of the vector is greater than the maximum “ z ” component, i.e. $\sqrt{j(j+1)} > j$. So you can’t get the angular momentum point perfectly along z direction. This is a bit suspicious right ! “Why can’t it so”. You might think that, all right why don’t I pick my axes such way that “ z ” points along the angular momentum vector $\hat{\mathbf{J}}$. Classically this is possible. But quantum mechanically we simply cannot. This is because in order to align with $\hat{\mathbf{J}}$ we need to know all the “three” components simultaneously. But “uncertainty principle” says you can’t. So if we know J_z then J_x and J_y are indeterminate.

We represent $\hat{\mathbf{J}}$ as follows; $\hat{\mathbf{J}}$ rotates along the surface of a cone about it’s axis. The cone’s height is equal to $m\hbar$. For $m = 0$ this is a circle. The tip of $\hat{\mathbf{J}}$ rotates in xy plane with a circle of radius $\hbar\sqrt{j(j+1)}$. The half angle of the cone is given by :

$$\theta_m = \arccos\left(\frac{m}{\sqrt{j(j+1)}}\right) \quad \text{with } m = j, j-1, \dots, -j+1, -j \quad (1.68)$$

So it rotates along the surface of a cone of half angle θ such that the projection along z axis is always $m\hbar$. Now as m can take $2j+1$ discrete values, so all θ values are not allowed. θ is “discrete and quantized”. The allowed values are :

$$\theta = \arccos\left(\frac{j}{\sqrt{j(j+1)}}\right), \arccos\left(\frac{j-1}{\sqrt{j(j+1)}}\right), \dots, \arccos\left(\frac{-j+1}{\sqrt{j(j+1)}}\right), \arccos\left(\frac{-j}{\sqrt{j(j+1)}}\right) \quad (1.69)$$

Since all the orientations of $\hat{\mathbf{J}}$ on the surface of the cone is equally likely, projections along x and y averages to zero, i.e.

$$\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0^{15} \quad (1.70)$$

To put it simple way, consider a sphere of radius $\hbar\sqrt{j(j+1)}$. Now the $\hat{\mathbf{J}}$ can make some certain discrete angles with z axis given by Eq. (1.69). To represent this what we can do is take a plane which is parallel to xy plane and that cuts z axis at a height $m\hbar$. Now this plane will cut the sphere along a path which can be obtained taking $\theta = \cos^{-1}(m/\sqrt{j(j+1)})$ and changing ϕ value. The tip of $\hat{\mathbf{J}}$ will always lie along this path ; which is basically the surface of the cone with half angle θ_m , that rotates w.r.t z axis. This is shown in this Fig (1.2). Again if we take a cross section of this along a plane passing through z axis¹⁶ then this looks like tip of $\hat{\mathbf{J}}$ lies on a circle of radius $\hbar\sqrt{j(j+1)}$ with certain θ values.

Problem 1.19

Find the angle between $\hat{\mathbf{J}}$ and ‘ z ’ axis for all possible orientations for $j = 4$.

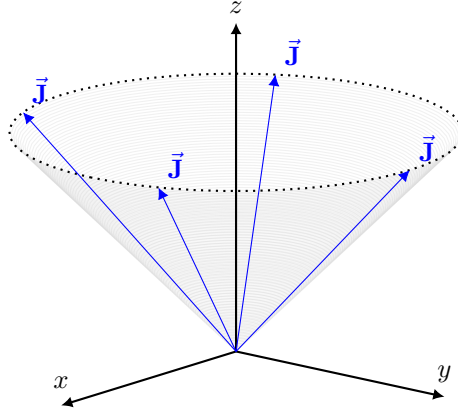
1.7.1 Representation for $j = \frac{1}{2}$

For $j = \frac{1}{2}$ we have $m = \pm\frac{1}{2}$. Using Eq. (1.69) we have the θ values as :

$$\theta_{\pm} = \arccos\left(\frac{\pm\frac{1}{2}}{\sqrt{\frac{1}{2}(\frac{1}{2}+1)}}\right) = \arccos\left(\pm\frac{1}{\sqrt{3}}\right) = \pm 54.7356^\circ \quad (1.71)$$

¹⁵We have computed it before in **Problem (1.13)**.

¹⁶ ϕ = constant surface

Figure 1.2: Geometric representation of $\vec{J}$.

This is shown graphically in Fig (1.3).

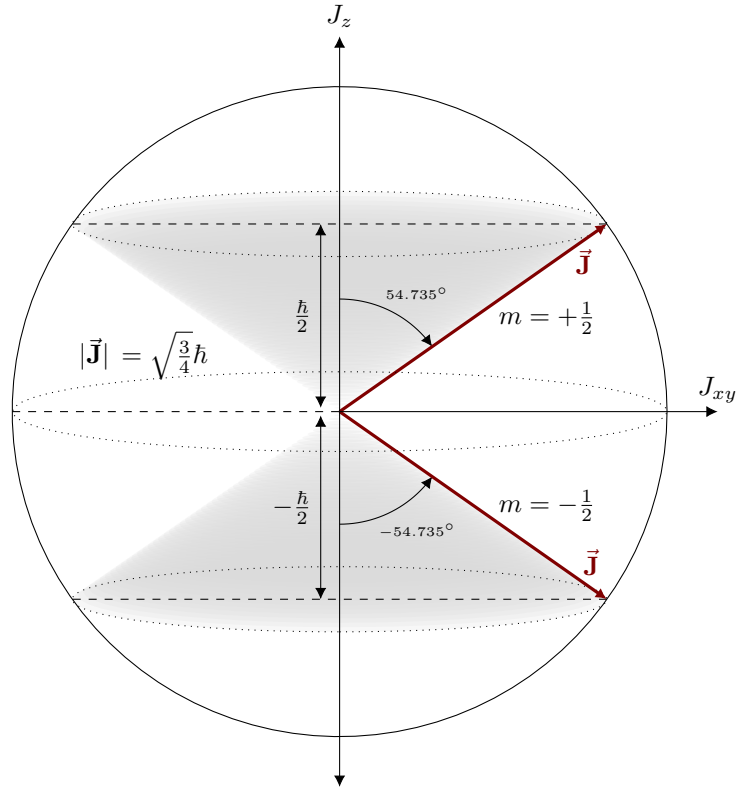


Figure 1.3: Geometric representation for $j = \frac{1}{2}$. The radius of the sphere is $|\vec{J}| = \sqrt{\frac{3}{4}}\hbar$ and only three projections are allowed, $m = \pm\frac{1}{2}$.

1.7.2 Representation for $j = 1$

For $j = 1$ we have $m = +1, 0, -1$. Using Eq. (1.69) we have the θ values as :

$$\theta = \arccos\left(\frac{1}{\sqrt{2}}\right), \arccos(0), \arccos\left(-\frac{1}{\sqrt{2}}\right) = 45^\circ, 90^\circ, 135^\circ \quad (1.72)$$

This is shown graphically in Fig (1.4).

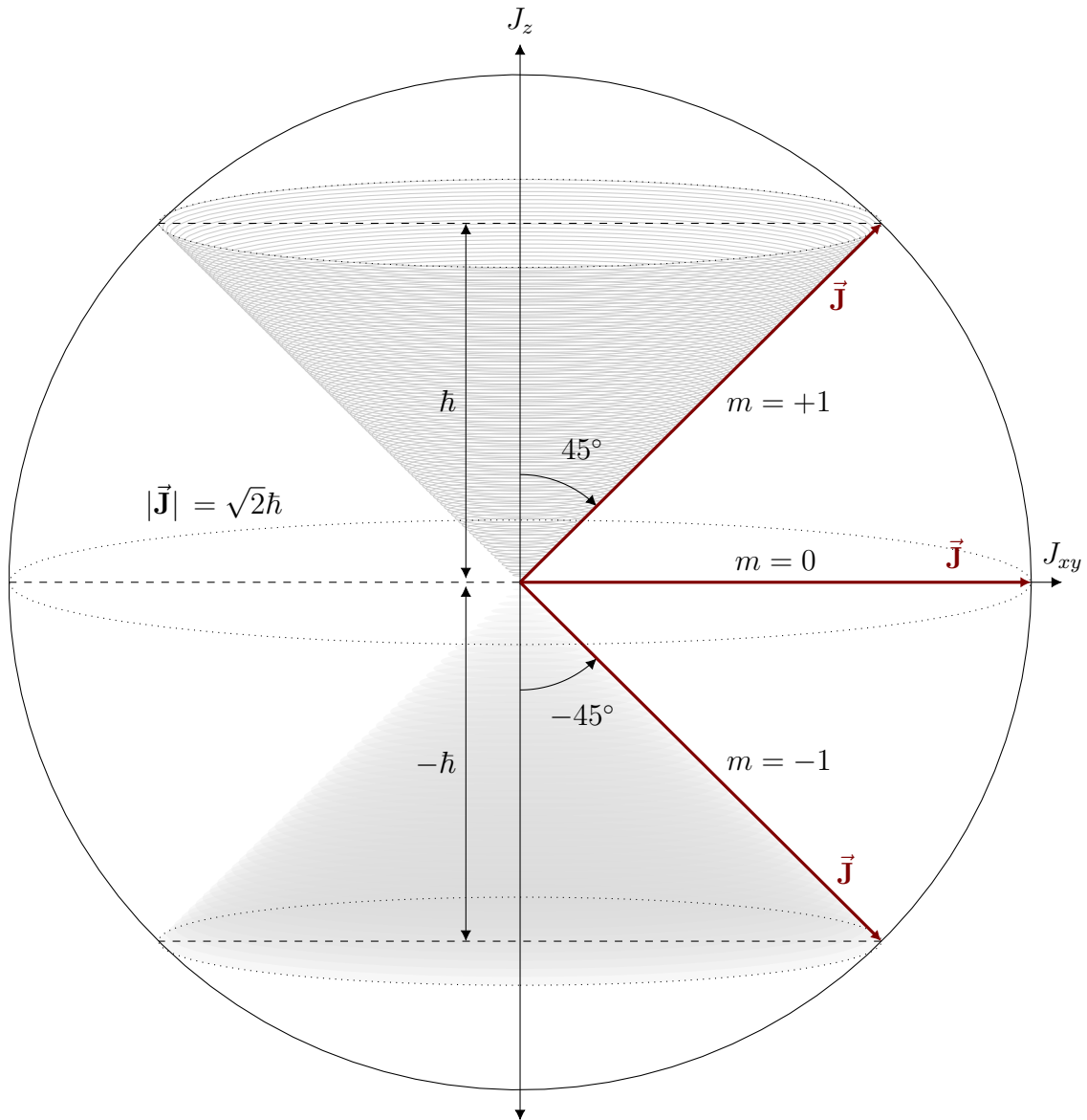


Figure 1.4: Geometric representation for $j = 1$. The radius of the sphere is $|\vec{J}| = \sqrt{2}\hbar$ and only three projections are allowed, $m = +1, 0, -1$.

1.8 Eigen value and Eigen functions of orbital angular momentum

We have talked about the matrix representation of the operators and eigen states. In this section we will try to find out their representation in position basis. We will mainly work in spherical polar coordinate system. Denoting joint eigenstates of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$ by $|\ell, m\rangle$, we have the eigenvalue equations :

$$\hat{\mathbf{L}}^2 |\ell, m\rangle = \ell(\ell + 1)\hbar^2 |\ell, m\rangle \quad (1.73a)$$

$$\hat{L}_z |\ell, m\rangle = m\hbar |\ell, m\rangle \quad (1.73b)$$

In position basis representation:

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \quad (1.74a)$$

$$\hat{\mathbf{L}}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \quad (1.74b)$$

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y \quad (1.74c)$$

$$= \pm \hbar e^{\pm i\phi} \left[\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right] \quad (1.74d)$$

As $\hat{L}_z$ and $\hat{\mathbf{L}}^2$ depend only on θ and ϕ , we can write the eigenfunction in the position basis representation as $f_{\ell}^m(\theta, \phi)$. i.e.

$$f_{\ell}^m(\theta, \phi) = \langle \theta, \phi | \ell, m \rangle \quad (1.75)$$

Here $f_{\ell}^m(\theta, \phi)$ is a continuous function of θ and ϕ and we wish to find the form of $f_{\ell}^m(\theta, \phi)$. Now in position basis representation Eq. (1.73) becomes :

$$\hat{\mathbf{L}}^2 f_{\ell}^m(\theta, \phi) = \ell(\ell + 1)\hbar^2 f_{\ell}^m(\theta, \phi) \quad (1.76a)$$

$$\hat{L}_z f_{\ell}^m(\theta, \phi) = m\hbar f_{\ell}^m(\theta, \phi) \quad (1.76b)$$

Where the form of operators $\hat{L}_z$ and $\hat{\mathbf{L}}^2$ are given by Eq. (1.74). All that is left is to solve the PDE's in order to obtain $f_{\ell}^m(\theta, \phi)$.

We also infer that :

$$\hat{L}_{\pm} f_{\ell}^m(\theta, \phi) = \hbar \sqrt{\ell(\ell + 1) - m(m \pm 1)} f_{\ell}^{m \pm 1}(\theta, \phi) \quad (1.77)$$

Which we is done explicitly in ???. In order to solve the PDE we use separation of variables, the technique we use in most of the cases.

$$f_{\ell}^m(\theta, \phi) = \Theta_{\ell m}(\theta) \Phi_m(\phi)^{17} \quad (1.78)$$

Using the form of $f_{\ell}^m(\theta, \phi)$ as given by Eq. (1.78) and using Eq. (1.74a) and Eq. (1.76b) we get :

$$\begin{aligned} -i\hbar \frac{\partial}{\partial \phi} [\Theta_{\ell m}(\theta) \Phi_m(\phi)] &= m\hbar \Theta_{\ell m}(\theta) \Phi_m(\phi) \\ \text{or } -i \frac{d\Phi_m(\phi)}{d\phi} &= m \Phi_m(\phi) \\ \text{or } \frac{d\Phi_m(\phi)}{\Phi_m(\phi)} &= im d\phi \end{aligned}$$

Integrating which gives :

$$\Phi_m(\phi) = C e^{im\phi} \quad (1.79)$$

¹⁷You might be wondering why I have put the subscript 'm' in $\Phi_m(\phi)$ and ' ℓm ' in $\Theta_{\ell m}(\theta)$. For now all I could say, hold on for a bit as it will be very clear in a couple of moments.

The constant C is found out by the appropriate normalization, i.e.

$$\begin{aligned}\int_0^\pi |\Phi_m(\phi)|^2 d\phi &= 1 \\ C^2 \int_0^{2\pi} d\phi &= 1 \\ C &= \frac{1}{\sqrt{2\pi}}\end{aligned}$$

So the normalized solution is given by :

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (1.80)$$

Problem 1.20 Orthonormality of $\Phi_m(\phi)$

Prove the orthonormality condition stated below :

$$\int_0^{2\pi} [\Phi_{m'}(\phi)]^* \Phi_m(\phi) d\phi = \delta_{m,m'} \quad (1.81)$$

Hint : Take the form of $\Phi_m(\phi)$, as given in Eq. (1.80) and perform the integration for two cases, $m = m'$ and $m \neq m'$ separately.

Now we enforce one condition that $\Phi_m(\phi)$ must be single valued. By this we mean that suppose we advance ϕ by 2π amount. Then it should return to the same value. i.e. $\Phi_m(\phi)$ is periodic with a period of 2π , which gives :

$$\Phi_m(\phi + 2\pi) = \Phi_m(\phi) \quad (1.82)$$

Or in other words we must have, $e^{2\pi im} = 1$. Which gives $m = 0, \pm 1, \pm 2, \dots$

This condition allows m values to be integers only ($m \in \mathbb{Z}$). Whereas in the previous section where we described the generic theory, we had restrictions on m to be either integer or a half-integer. This is a consequence of orbital angular momentum. m can take integer values if and only if ℓ is a non negative integer, i.e. $\ell \in \mathbb{Z}_{\geq 0}$. So far we got the restrictions on the nature of eigenvalues of operators $\hat{L}^2$ and $\hat{L}_z$ (i.e. ℓ and m). The only thing left is to solve for $\Theta_{\ell m}(\theta)$.

Using the form of $\Phi_m(\phi)$ as given by Eq. (1.80) and combining Eq. (1.74b) and Eq. (1.76a) we got :

$$\begin{aligned}\hat{L}^2 f_\ell^m(\theta, \phi) &= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \frac{\Theta_{\ell m}(\theta)}{\sqrt{2\pi}} e^{im\phi} \\ &= -\frac{\hbar^2}{\sqrt{2\pi}} e^{im\phi} \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) + \frac{i^2 m^2}{\sin^2 \theta} \right] \Theta_{\ell m}(\theta) \\ &= -\hbar^2 \Phi_m(\phi) \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \right] \Theta_{\ell m}(\theta) \\ &= \ell(\ell + 1) \hbar^2 \Theta_{\ell m}(\theta) \Phi_m(\phi)\end{aligned}$$

Which gives a 2nd order differential equation for $\Theta_{\ell m}(\theta)$, which is given by :

$$\begin{aligned}\left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \right] \Theta_{\ell m}(\theta) &= -\ell(\ell + 1) \Theta_{\ell m}(\theta) \\ \text{or } \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta_{\ell m}(\theta)}{d\theta} \right) + \left[\ell(\ell + 1) - \frac{m^2}{\sin^2 \theta} \right] \Theta_{\ell m}(\theta) &= 0\end{aligned} \quad (1.83)$$

Eq. (1.83) is known as *general Legendre differential equation*, whose solutions are *associated Legendre polynomials* $P_\ell^m(\cos \theta)$ given by :

$$\Theta_{\ell m}(\theta) = C_{\ell m} P_\ell^m(\cos \theta) \quad (1.84)$$

$C_{\ell m}$ is the appropriate normalization constant. Now **associated Legendre polynomials** are defined as :

$$P_{\ell}^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_{\ell}(x), \quad \text{for } m \geq 0^{18} \quad (1.85)$$

For negative values of m we define :

$$P_{\ell}^{-m}(x) = (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_{\ell}^m(x), \quad \text{where } m \geq 0 \quad (1.86)$$

$P_{\ell}(x)$ is the ℓ^{th} order **Legendre polynomial**, defined by Rodrigues' formula :

$$P_{\ell}(x) = \frac{1}{2^{\ell} \ell!} \frac{d^{\ell}}{dx^{\ell}} \left[x^2 - 1 \right]^{\ell} \quad (1.87)$$

Using Eq. (1.85), Eq. (1.86) and Eq. (1.87) we can easily see :

$$P_{\ell}(-x) = (-1)^{\ell} P_{\ell}(x) \quad (1.88a)$$

$$P_{\ell}^m(-x) = (-1)^{\ell+m} P_{\ell}^m(x)^{19} \quad (1.88b)$$

Problem 1.21 Legendre Polynomials

Using the Rodrigues' formula Eq. (1.87) show that :

$$P_0(x) = 1 \quad (1.89a)$$

$$P_1(x) = x \quad (1.89b)$$

$$P_2(x) = \frac{3x^2 - 1}{2} \quad (1.89c)$$

$$P_3(x) = \frac{5x^3 - 3x}{2} \quad (1.89d)$$

$$P_4(x) = \frac{35x^4 - 30x^2 + 3}{8} \quad (1.89e)$$

$$P_5(x) = \frac{63x^5 - 70x^3 + 15x}{8} \quad (1.89f)$$

Also plot them as a function of x in the range $[-1, 1]$.

Problem 1.22 Associated Legendre functions

Using the definition given by Eq. (1.85) and the results obtained from Problem 1.2 show that :

$$P_1^1(x) = -\sqrt{1-x^2} \quad (1.90a)$$

$$P_2^1(x) = -3x\sqrt{1-x^2} \quad (1.90b)$$

$$P_2^2(x) = 3(1-x^2) \quad (1.90c)$$

$$P_3^1(x) = -\frac{3}{2}(5x^2-1)\sqrt{1-x^2} \quad (1.90d)$$

$$P_3^2(x) = 15x(1-x^2) \quad (1.90e)$$

$$P_3^3(x) = -15(1-x^2)^{3/2} \quad (1.90f)$$

Now substitute $x = \cos \theta$ in those equations and rewrite them as a function of θ .

Also draw polar plot of them, where θ is polar angle. It varies in range $[0, \pi]$. Take the radius as $r = |P_{\ell}^m(\cos \theta)|$.

¹⁸The factor $(-1)^m$ is also known as *Condon-Shortley* phase, which is merely a convention, nothing special.

¹⁹Here m can be negative also.

Problem 1.23 Parity of P_ℓ

Prove the relation given in Eq. (1.88a)

Hint : Start with Eq. (1.87) with the substitution $y = -x$, which gives $dy = -dx$.

So the operator $\frac{d}{dx}$ becomes $\frac{d}{dx} = \frac{dy}{dx} \times \frac{d}{dy} = -\frac{d}{dy}$. Implied :

$$\left(\frac{d}{dx}\right)^\ell \equiv (-1)^\ell \left(\frac{d}{dy}\right)^\ell$$

I am sure you can do the rest.

Example 1.7 Parity of P_ℓ^m

Prove the relation as given in Eq. (1.88b), which holds for any integer m , where $-\ell \leq m \leq \ell$.

Ans: Using Eq. (1.85) and Eq. (1.88a) we have :

$$\begin{aligned} P_\ell^m(-x) &= (-1)^m [1 - (-x)^2]^{m/2} \left(\frac{d}{d(-x)}\right)^m P_\ell(-x) \\ &= (-1)^m \left[(-1)^m (1 - x^2)^{m/2} \left(\frac{d}{dx}\right)^m (-1)^\ell P_\ell(x) \right] \\ &= (-1)^{\ell+m} \left[(-1)^m (1 - x^2)^{m/2} \left(\frac{d}{dx}\right)^m P_\ell(x) \right] \\ &= (-1)^{\ell+m} P_\ell^m(x) \end{aligned}$$

The thing we proved is for $m \geq 0$ as the functional for defined in Eq. (1.85) is valid for those m values only. Now for negative values of m we use Eq. (1.86).

$$\begin{aligned} P_\ell^{-m}(-x) &= (-1)^m \frac{(\ell - m)!}{(\ell + m)!} P_\ell^m(-x) \\ &= (-1)^m \frac{(\ell - m)!}{(\ell + m)!} [(-1)^{\ell+m} P_\ell^m(x)] \\ &= (-1)^{m+\ell+m} \frac{(\ell - m)!}{(\ell + m)!} P_\ell^m(x) \\ &= (-1)^{\ell+m} \left[(-1)^m \frac{(\ell - m)!}{(\ell + m)!} P_\ell^m(x) \right] \\ &= (-1)^{\ell+m} P_\ell^{-m}(x) \\ &= (-1)^{\ell-m} P_\ell^{-m}(x) \end{aligned}$$

Where in the last line we have used the fact that $(-1)^{-2m} = 1$. Q.E.D.

Problem 1.24 Derivative of $P_\ell^m(x)$

Using the defining formula Eq. (1.85), prove that

$$(1 - x^2) \frac{dP_\ell^m}{dx} = -\sqrt{1 - x^2} P_\ell^{m+1} - mx P_\ell^m \quad (1.91)$$

Hint: Take derivative on both sides of Eq. (1.85).

$$\begin{aligned}
 \frac{dP_\ell^m}{dx} &= -2x \frac{m}{2} (-1)^m (1-x^2)^{\frac{m}{2}-1} \frac{d^m}{dx^m} P_\ell(x) + (-1)^m (1-x^2)^{m/2} \frac{d^{m+1}}{dx^{m+1}} P_\ell(x) \\
 &= -\frac{mx}{1-x^2} P_\ell^m - \frac{1}{\sqrt{1-x^2}} (-1)^{m+1} (1-x^2)^{(m+1)/2} \frac{d^{m+1}}{dx^{m+1}} P_\ell(x) \\
 &= -\frac{mx}{1-x^2} P_\ell^m - \frac{1}{\sqrt{1-x^2}} P_\ell^{m+1}
 \end{aligned} \tag{1.92}$$

Note that this is valid only for $m \geq 0$.

Now for negative values of m we cannot use this equation. Taking derivative on both sides of Eq. (1.86) we get :

$$\begin{aligned}
 \frac{dP_\ell^{-m}}{dx} &= (-1)^m \frac{(\ell-m)!}{(\ell+m)!} \frac{dP_\ell^m}{dx} \\
 &= (-1)^m \frac{(\ell-m)!}{(\ell+m)!} \left[-\frac{mx}{1-x^2} P_\ell^m - \frac{1}{\sqrt{1-x^2}} P_\ell^{m+1} \right] \\
 &= -\frac{mx}{1-x^2} \left[(-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m \right] - \frac{1}{\sqrt{1-x^2}} \left[(-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^{m+1} \right] \\
 &= -\frac{mx}{1-x^2} \left[(-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m \right] - \frac{1}{\sqrt{1-x^2}} \left[(-1)^m \frac{(\ell-m)!}{(\ell+m)!} \times (-1)^{-(m+1)} \frac{(\ell+m+1)!}{(\ell-m-1)!} P_\ell^{-(m+1)} \right]^{20} \\
 &= -\frac{mx}{1-x^2} \left[(-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m \right] - \frac{1}{\sqrt{1-x^2}} \left[(-1)^{(m-m-1)} (\ell+m+1)(\ell-m) P_\ell^{-(m+1)} \right] \\
 &= -\frac{mx}{1-x^2} P_\ell^{-m} + \frac{(\ell+m+1)(\ell-m)}{\sqrt{1-x^2}} P_\ell^{-(m+1)}
 \end{aligned} \tag{1.93}$$

Problem 1.25

Take $\ell = 2$. Now using Eq. (1.92) and Eq. (1.93) compute the L.H.S. and R.H.S separately and then compare whether these relations hold or not. For example calculate the derivatives of $\frac{dP_2^0}{dx}$, $\frac{dP_2^1}{dx}$, $\frac{dP_2^2}{dx}$, $\frac{dP_2^{-1}}{dx}$, $\frac{dP_2^{-2}}{dx}$ and also evaluate the R.H.S.

1.8.1 Orthonormality

By definition :

$$\langle \ell', m' | \ell, m \rangle = \delta_{\ell, \ell'} \delta_{m, m'} \tag{1.94}$$

Now in order to represent this in the position basis we use the completeness relation in $\langle \theta, \phi |$ basis which is given by :

$$\int d\Omega^{21} |\theta, \phi\rangle \langle \theta, \phi| = 1 \tag{1.95}$$

Now combining Eq. (1.94) and Eq. (1.95) we get :

$$\begin{aligned}
 \delta_{\ell, \ell'} \delta_{m, m'} &= \langle \ell', m' | \left[\int d\Omega |\theta, \phi\rangle \langle \theta, \phi| \right] | \ell, m \rangle \\
 &= \int d\Omega \langle \ell', m' | \theta, \phi \rangle \langle \theta, \phi | \ell, m \rangle \\
 &= \int d\Omega \left[f_{\ell'}^{m'}(\theta, \phi) \right]^* f_\ell^m(\theta, \phi)^{22}
 \end{aligned} \tag{1.96}$$

This is the orthonormality condition in position basis representation. So far we have made some progress in finding the form of $f_\ell^m(\theta, \phi)$. The only thing that is left is to solve for the normalization constant $C_{\ell m}$ appeared in Eq. (1.83). Using

²¹Where $\int d\Omega = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi$

²²Where we have used Eq. (1.75) and $\langle \ell, m | \theta, \phi \rangle = [\langle \theta, \phi | \ell, m \rangle]^\dagger = [f_\ell^m(\theta, \phi)]^*$

Eq. (1.78) and Eq. (1.84) in Eq. (1.96) we have :

$$\begin{aligned}
 \delta_{\ell,\ell'} \delta_{m,m'} &= \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi [\Theta_{\ell',m'}(\theta) \Phi_{m'}(\phi)]^* \Theta_{\ell m}(\theta) \Phi_m(\phi) \\
 &= \int_0^\pi \sin \theta d\theta [\Theta_{\ell',m'}(\theta)]^* \Theta_{\ell m}(\theta) \underbrace{\int_0^{2\pi} d\phi [\Phi_{m'}(\phi)]^* \Phi_m(\phi)}_{=\delta_{m,m'} \text{ Using Eq. (1.81)}} \\
 &= \int_0^\pi \sin \theta d\theta [C_{\ell',m'} P_{\ell'}^{m'}(\cos \theta)]^* C_{\ell m} P_\ell^m(\cos \theta) \delta_{m,m'} \\
 &= \delta_{m,m'} C_{\ell',m}^* C_{\ell m} \int_0^\pi \sin \theta d\theta P_{\ell'}^m(\cos \theta) P_\ell^m(\cos \theta) \\
 &= \frac{2}{2\ell+1} \frac{(\ell+m)!}{(\ell-m)!} \delta_{\ell,\ell'}^{23} \delta_{m,m'} C_{\ell',m}^* C_{\ell m} \\
 &= \frac{2}{2\ell+1} \frac{(\ell+m)!}{(\ell-m)!} \delta_{\ell,\ell'} \delta_{m,m'} |C_{\ell m}|^2
 \end{aligned}$$

Which gives us the value of the normalization constant as :

$$C_{\ell m} = \sqrt{\frac{2\ell+1}{2} \frac{(\ell-m)!}{(\ell+m)!}} \quad (1.98)$$

Using Eq. (1.78), Eq. (1.80), Eq. (1.84) and Eq. (1.98) we finally have the form of $f_\ell^m(\theta, \phi)$ as follows :

$$f_\ell^m(\theta, \phi) = Y_\ell^m(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\phi} \quad (1.99)$$

The solution which we call as $Y_\ell^m(\theta, \phi)$ is widely known as **Spherical Harmonics**.

Example 1.8 Property of Spherical Harmonics

Prove that :

$$Y_\ell^{-m}(\theta, \phi) = (-1)^m [Y_\ell^m(\theta, \phi)]^* \quad \text{where } m \geq 0 \quad (1.100)$$

Ans: Using Eq. (1.99) we have :

$$\begin{aligned}
 Y_\ell^{-m}(\theta, \phi) &= \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell+m)!}{(\ell-m)!}} P_\ell^{-m}(\cos \theta) e^{-im\phi} \\
 &= \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell+m)!}{(\ell-m)!}} (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m(\cos \theta) e^{-im\phi} \\
 &= (-1)^m \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{-im\phi} \\
 &= (-1)^m \left[\sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\phi} \right]^* \\
 &= (-1)^m [Y_\ell^m(\theta, \phi)]^* \text{ (QED)}
 \end{aligned}$$

²³The orthogonality of the associated Legendre functions are given by :

$$\int_0^\pi \sin \theta d\theta P_{\ell'}^m(\cos \theta) P_\ell^m(\cos \theta) = \frac{2}{2\ell+1} \frac{(\ell+m)!}{(\ell-m)!} \delta_{\ell,\ell'} \quad (1.97)$$

Problem 1.26 Spherical Harmonics

Using the definitions show that

$$Y_0^0(\theta, \phi) = \frac{1}{\sqrt{4\pi}} \quad (1.101a)$$

$$Y_1^0(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta \quad (1.101b)$$

$$Y_1^{\pm 1}(\theta, \phi) = \mp \left(\frac{3}{8\pi}\right)^{1/2} \sin \theta e^{\pm i\phi} \quad (1.101c)$$

$$Y_2^0(\theta, \phi) = \left(\frac{5}{16\pi}\right)^{1/2} (3 \cos^2 \theta - 1) \quad (1.101d)$$

$$Y_2^{\pm 1}(\theta, \phi) = \mp \left(\frac{15}{8\pi}\right)^{1/2} \sin \theta \cos \theta e^{\pm i\phi} \quad (1.101e)$$

$$Y_2^{\pm 2}(\theta, \phi) = \left(\frac{15}{32\pi}\right)^{1/2} \sin^2 \theta e^{\pm 2i\phi} \quad (1.101f)$$

Using the appropriate substitutions write them in terms of (x, y, z) . i.e. substitute

$$\cos \theta = \frac{z}{r}, \quad \sin \theta \cos \phi = \frac{x}{r}, \quad \sin \theta \sin \phi = \frac{y}{r}. \quad (1.102)$$

Example 1.9

For $m = 0$, the spherical harmonics are given by :

$$Y_\ell^{m=0}(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi}} P_\ell(\cos \theta)$$

(a). Write down the expression of $Y_3^0(\theta, \phi)$.

Ans.

$$Y_3^0(\theta, \phi) = \sqrt{\frac{2 \cdot 3 + 1}{4\pi}} P_3(\cos \theta)$$

Using $P_3(\cos \theta) = \frac{1}{2}(5 \cos^3 \theta - 3 \cos \theta)$ we get :

$$Y_3^0(\theta, \phi) = \sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$$

(b). Write it in Cartesian coordinates.

Ans.

Putting $\cos \theta = \frac{z}{r}$ we get :

$$Y_3^0(x, y, z) = \sqrt{\frac{7}{16\pi}} \left(\frac{5z^3}{r^3} - \frac{3z}{r} \right)$$

(c). Using $\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$, Find out $\hat{L}_z Y_3^0(x, y, z)$.

Ans.

We compute :

$$x \frac{\partial}{\partial y} \left(\frac{5z^3}{r^3} - \frac{3z}{r} \right) = x \left[-\frac{15z^3}{r^4} \frac{\partial r}{\partial y} + \frac{3z}{r^2} \frac{\partial r}{\partial y} \right] = -\frac{15xyz^3}{r^5} + \frac{3xyz}{r^3}$$

Similarly,

$$y \frac{\partial}{\partial x} \left(\frac{5z^3}{r^3} - \frac{3z}{r} \right) = y \left[-\frac{15z^3}{r^4} \frac{\partial r}{\partial x} + \frac{3z}{r^2} \frac{\partial r}{\partial x} \right] = -\frac{15xyz^3}{r^5} + \frac{3xyz}{r^3}$$

Subtracting we get :

$$\hat{L}_z Y_3^0(x, y, z) = 0$$

(d). For $m = 0$, we can write:

$$f_\ell^{m=0}(\theta, \phi) = \frac{C_{\ell 0}}{\sqrt{2\pi}} P_\ell(\cos \theta)$$

Take $\ell = 3$ and using the form of $P_3(\cos \theta)$, normalize $f_\ell^{m=0}(\theta, \phi)$, i.e., find out the value C_{30} . Check with the stated normalization: $\frac{C_{30}}{\sqrt{2\pi}} = \sqrt{\frac{7}{4\pi}}$ (should be).

Normalization

$$\begin{aligned} \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi |f_\ell^m|^2 &= 1 \\ \Rightarrow \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \cdot \frac{|C_{\ell 0}|^2}{2\pi} \cdot \frac{(5\cos^3 \theta - 3\cos \theta)^2}{4} &= 1 \\ \Rightarrow \int_0^\pi \sin \theta d\theta \frac{(5\cos^3 \theta - 3\cos \theta)^2}{4} |C_{\ell 0}|^2 &= 1 \end{aligned}$$

Take $\xi = \cos \theta$, then $d\xi = -\sin \theta d\theta$. Limits: for $\theta = 0, \xi = 1$ and for $\theta = \pi, \xi = -1$.

$$\begin{aligned} \Rightarrow \int_1^{-1} -d\xi \frac{(5\xi^3 - 3\xi)^2}{4} |C_{\ell 0}|^2 &= 1 \\ \Rightarrow C_{\ell 0} &= \frac{1}{\sqrt{\int_{-1}^1 d\xi \frac{(5\xi^3 - 3\xi)^2}{4}}} = \frac{1}{\sqrt{\int_{-1}^1 \frac{d\xi}{4} (25\xi^6 - 30\xi^4 + 9\xi^2)}} \\ &= \frac{1}{\sqrt{\frac{2}{4} \left[\frac{25\xi^7}{7} - \frac{30\xi^5}{5} + \frac{9\xi^3}{3} \right]_0^1}} = \frac{1}{\sqrt{\frac{2}{4} \left(\frac{25}{7} - 6 + 3 \right)}} \\ &= \frac{1}{\sqrt{\frac{2}{4} \left(\frac{25}{7} - 3 \right)}} = \frac{1}{\sqrt{\frac{2}{4} \cdot \frac{4}{7}}} = \sqrt{\frac{7}{2}} \end{aligned}$$

$\Rightarrow C_{\ell 0} = \sqrt{\frac{7}{2}}$. Thus, $\frac{C_{\ell 0}}{\sqrt{2\pi}} = \sqrt{\frac{7}{4\pi}}$ (Got it!).

(e). Find out the eigenvalue of $\hat{\mathbf{L}}^2$ operator for this state.

Ans: We already know for $\ell = 3$, the eigenvalue of $\hat{\mathbf{L}}^2$ is $\ell(\ell + 1)\hbar^2 = 12\hbar^2$. Now, from the eigenfunction, we will show it directly.

$$\begin{aligned} \nabla^2 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{\mathbf{L}}^2}{\hbar^2 r^2} \end{aligned}$$

Now :

$$\nabla^2 Y_3^0(\theta, \phi) = \frac{1}{\hbar^2 r^2} \hat{\mathbf{L}}^2 Y_3^0(\theta, \phi) \Rightarrow \hat{\mathbf{L}}^2 Y_3^0(\theta, \phi) = -\hbar^2 r^2 \nabla^2 Y_3^0(\theta, \phi)$$

In Cartesian coordinates: $Y_3^0(x, y, z) = \sqrt{\frac{7}{16\pi}} \left(\frac{5z^3}{r^3} - \frac{3z}{r} \right)$. Using this we calculate :

$$\begin{aligned}\frac{\partial Y_3^0}{\partial x} &= \sqrt{\frac{7}{16\pi}} \left(-\frac{15z^3x}{r^5} + \frac{3zx}{r^3} \right) \\ \frac{\partial^2 Y_3^0}{\partial x^2} &= \sqrt{\frac{7}{16\pi}} \left(-\frac{15z^3}{r^5} + \frac{3z}{r^3} + \frac{75z^3x^2}{r^7} - \frac{9zx^2}{r^5} \right)\end{aligned}$$

Replacing $x \rightarrow y$ gives :

$$\frac{\partial^2 Y_3^0}{\partial y^2} = \sqrt{\frac{7}{16\pi}} \left(-\frac{15z^3}{r^5} + \frac{3z}{r^3} + \frac{75z^3y^2}{r^7} - \frac{9zy^2}{r^5} \right)$$

Again for z we compute :

$$\begin{aligned}\frac{\partial Y_3^0}{\partial z} &= \sqrt{\frac{7}{16\pi}} \left(-\frac{15z^4}{r^5} + \frac{3z^2}{r^3} - \frac{3}{r} + \frac{15z^2}{r^3} \right) \\ \frac{\partial^2 Y_3^0}{\partial z^2} &= \sqrt{\frac{7}{16\pi}} \left(-\frac{60z^3}{r^5} + \frac{6z}{r^3} + \frac{3z}{r^3} + \frac{30z}{r^3} + \frac{75z^5}{r^7} - \frac{9z^3}{r^5} - \frac{45z^3}{r^5} \right) \\ &= \sqrt{\frac{7}{16\pi}} \left(-\frac{114z^3}{r^5} + \frac{39z}{r^3} + \frac{75z^5}{r^7} \right)\end{aligned}$$

Summing these for $\nabla^2 Y_3^0(\theta, \phi)$:

$$\begin{aligned}\nabla^2 Y_3^0 &= \sqrt{\frac{7}{16\pi}} \left(\frac{36z}{r^3} - \frac{60z^3}{r^5} \right) \\ \Rightarrow \hat{\mathbf{L}}^2 Y_3^0 &= -\hbar^2 r^2 \left[\sqrt{\frac{7}{16\pi}} \left(\frac{36z}{r^3} - \frac{60z^3}{r^5} \right) \right] \\ &= 12\hbar^2 \sqrt{\frac{7}{16\pi}} \left(\frac{5z^3}{r^3} - \frac{3z}{r} \right) = 12\hbar^2 Y_3^0\end{aligned}$$

Therefore eigenvalue is $12\hbar^2$.

Another Way: Now calculate $\hat{\mathbf{L}}^2 Y_3^0(\theta, \phi)$.

$$\begin{aligned}
 \hat{\mathbf{L}}^2 Y_3^0(\theta, \phi) &= -\frac{\hbar^2}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{dY_3^0(\theta, \phi)}{d\theta} \right] \\
 &= -\frac{\hbar^2}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d}{d\theta} \left[\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta) \right] \right] \\
 &= -\frac{\hbar^2}{\sin \theta} \sqrt{\frac{7}{16\pi}} \frac{d}{d\theta} \left[\sin \theta \frac{d}{d\theta} (5 \cos^3 \theta - 3 \cos \theta) \right] \\
 &= -\frac{\hbar^2}{\sin \theta} \sqrt{\frac{7}{16\pi}} \frac{d}{d\theta} [\sin \theta (-15 \cos^2 \theta \sin \theta + 3 \sin \theta)] \\
 &= -\frac{\hbar^2}{\sin \theta} \sqrt{\frac{7}{16\pi}} \frac{d}{d\theta} [-15 \cos^2 \theta \sin^2 \theta + 3 \sin^2 \theta] \\
 &= -\frac{\hbar^2}{\sin \theta} \sqrt{\frac{7}{16\pi}} [30 \cos \theta \sin^3 \theta - 30 \cos^3 \theta \sin \theta + 6 \sin \theta \cos \theta] \\
 &= -\hbar^2 \sqrt{\frac{7}{16\pi}} [30 \cos \theta \sin^2 \theta - 30 \cos^3 \theta + 6 \cos \theta] \\
 &= -\hbar^2 \sqrt{\frac{7}{16\pi}} [30 \cos \theta - 30 \cos^3 \theta - 30 \cos^3 \theta + 6 \cos \theta] \\
 &= -\hbar^2 \sqrt{\frac{7}{16\pi}} [36 \cos \theta - 60 \cos^3 \theta] \\
 &= 12\hbar^2 \sqrt{\frac{7}{16\pi}} [5 \cos^3 \theta - 3 \cos \theta] \\
 &= 12\hbar^2 Y_3^0(\theta, \phi)
 \end{aligned}$$

Example 1.10 Parity operator Eigenvalue and Eigenstate

Prove that $Y_\ell^m(\theta, \phi)$ is an eigenstate of parity operator $\hat{P}$ with eigenvalue $(-1)^\ell$.

Proof : The parity operator $\hat{P}$ changes $\vec{r} \rightarrow -\vec{r}$ which does not affect the radial coordinate r but $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow \pi + \phi$.

Now we have to see how this affects $Y_\ell^m(\theta, \phi)$. The answer is $Y_\ell^m(\theta, \phi) \rightarrow Y_\ell^m(\pi - \theta, \pi + \phi)$. Again :

$$\begin{aligned}
 Y_\ell^m(\pi - \theta, \pi + \phi) &= \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m[\cos(\pi - \theta)] e^{im(\pi + \phi)} \\
 &= e^{im\pi} \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(-\cos \theta) e^{im\phi} \\
 &= (-1)^m \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} [(-1)^{\ell+m} P_\ell^m(\cos \theta)]^{24} e^{im\phi} \\
 &= (-1)^{\ell+2m} \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\phi} \\
 &= (-1)^\ell Y_\ell^m(\theta, \phi)
 \end{aligned}$$

So we see that under the parity operation $\hat{P}Y_\ell^m(\theta, \phi) = (-1)^\ell Y_\ell^m(\theta, \phi)$. Which proves the given statement. (QED)

²³We have used that

$$P_\ell^{-(m+1)} = (-1)^{(m+1)} \frac{(\ell-m-1)!}{(\ell+m+1)!} P_\ell^{m+1} \Rightarrow P_\ell^{m+1} = (-1)^{-(m+1)} \frac{(\ell+m+1)!}{(\ell-m-1)!} P_\ell^{-(m+1)}$$

²⁴Using Eq. (1.88b)

Example 1.11

Consider the system initially in the state

$$\psi(\theta, \phi) = \frac{1}{\sqrt{5}} Y_1^{-1}(\theta, \phi) + \sqrt{\frac{3}{5}} Y_1^0(\theta, \phi) + \frac{1}{\sqrt{5}} Y_1^1(\theta, \phi) \quad (1.103)$$

- (a) Find $\langle \psi | \hat{L}_+ | \psi \rangle$.
- (b) If L_z is measured then what are the probabilities?
- (c) If we measure the “z” component of angular momentum and get $\langle \hat{L}_z \rangle = -\hbar$ then calculate ΔL_x , ΔL_y and $\Delta L_x \Delta L_y$.

Ans. Writing

$$Y_1^{-1}(\theta, \phi) = |1, -1\rangle, \quad Y_1^0(\theta, \phi) = |1, 0\rangle, \quad Y_1^1(\theta, \phi) = |1, 1\rangle$$

We have :

$$|\psi\rangle = \frac{1}{\sqrt{5}} |1, -1\rangle + \sqrt{\frac{3}{5}} |1, 0\rangle + \frac{1}{\sqrt{5}} |1, 1\rangle \quad (1.104)$$

- (a) Using Eq. (1.77) we have $\hat{L}_+ |1, -1\rangle = \hbar\sqrt{2} |1, 0\rangle$, $\hat{L}_+ |1, 0\rangle = \hbar\sqrt{2} |1, 1\rangle$, $\hat{L}_+ |1, 1\rangle = 0$. So we have :

$$\hat{L}_+ |\psi\rangle = \hbar\sqrt{2} \left[\frac{1}{\sqrt{5}} |1, 0\rangle + \sqrt{\frac{3}{5}} |1, 1\rangle \right]$$

Which gives : $\langle \psi | \hat{L}_+ | \psi \rangle = \hbar\sqrt{2} \left(\frac{\sqrt{3}}{5} + \frac{\sqrt{3}}{5} \right) = \frac{2\sqrt{6}}{5} \hbar$.

- (b) The corresponding probabilities are $P[\langle \hat{L}_z \rangle = -\hbar] = \frac{1}{5}$, $P[\langle \hat{L}_z \rangle = \hbar] = \frac{1}{5}$, $P[\langle \hat{L}_z \rangle = 0] = \frac{3}{5}$.

- (c) if $\langle \hat{L}_z \rangle = -\hbar$ then the state is $|\psi\rangle = |1, -1\rangle$. For a given state $|\ell, m\rangle$ we have $\Delta L_x = \Delta L_y = \sqrt{\frac{\hbar^2}{2} [\ell(\ell+1) - m^2]}$. Now $\ell = 1, m = -1$ gives : $\Delta L_x = \Delta L_y = \frac{\hbar}{\sqrt{2}}$. And also $\Delta L_x \Delta L_y = \frac{\hbar^2}{2} = \frac{\hbar}{2} |\langle \hat{L}_z \rangle|$.

Problem 1.27

Consider a state

$$\psi(\theta, \phi) = \frac{1}{\sqrt{14}} Y_1^1(\theta, \phi) + \frac{1}{\sqrt{8}} Y_1^0(\theta, \phi) + A Y_1^{-1}(\theta, \phi) \quad (1.105)$$

- (a) Find the constant A so that $\psi(\theta, \phi)$ is normalized.
- (b) Find $\hat{L}_\pm \psi(\theta, \phi)$.
- (c) Find $\langle \hat{L}_z \rangle$ and $\langle \hat{L}^2 \rangle$.
- (d) Now consider the following state :

$$\tilde{\psi}(\theta, \phi) = \sqrt{\frac{8}{15}} Y_1^1(\theta, \phi) + \sqrt{\frac{4}{15}} Y_1^0(\theta, \phi) + \sqrt{\frac{3}{15}} Y_2^{-1}(\theta, \phi) \quad (1.106)$$

Evaluate $\langle \tilde{\psi} | \hat{L}_z | \psi \rangle$ and $\langle \tilde{\psi} | \hat{L}_\pm | \psi \rangle$

Problem 1.28

1. Find out $\hat{L}_\pm \hat{L}_\mp Y_\ell^m(\theta, \phi)$ using Eq. (1.77).
2. Using the results obtained find out $[\hat{L}_+ \hat{L}_-, \hat{L}_- \hat{L}_+] Y_\ell^m(\theta, \phi)$.
3. Find out the following “anti-commutator” acting on $Y_\ell^m(\theta, \phi)$, $\{\hat{L}_+ \hat{L}_-, \hat{L}_- \hat{L}_+\} Y_\ell^m(\theta, \phi)$.

Problem 1.29

Consider a wave function $\psi(x, y, z)$ describing the state of a “quantum particle” :

$$\psi(x, y, z) = \frac{1}{4\sqrt{\pi}} \frac{z}{r} + \frac{1}{\sqrt{3\pi}} \frac{x}{r} \quad (1.107)$$

- (a) Write down $\psi(x, y, z)$ in terms of spherical harmonics.
- (b) Draw a polar plot of ψ .
- (c) Is this state normalized ?
- (d) Calculate $\hat{\mathbf{L}}^2 \psi$ and $\hat{L}_z \psi$.
- (e) Is ψ is an eigenstate of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$? If so, what are the corresponding eigenvalues.
- (f) Calculate $\hat{L}_\pm \psi$ and obtain $\langle \psi | \hat{L}_\pm | \psi \rangle$

Example 1.12 Finding the form of $Y_\ell^\ell(\theta, \phi)$

Using Eq. (1.99), show that

$$Y_\ell^\ell(\theta, \phi) = \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} \left[e^{i\phi} \sin \theta \right]^\ell \quad (1.108)$$

Ans : Using the form of $Y_\ell^m(\theta, \phi)$ as given by Eq. (1.99), and putting $m = \ell$ we get :

$$Y_\ell^\ell(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi}} \frac{1}{\sqrt{(2\ell)!}} e^{i\ell\phi} P_\ell^\ell(\cos \theta) \quad (1.109)$$

We need to find out what is $P_\ell^\ell(x)$. Using Eq. (1.85) and Eq. (1.87) we have :

$$P_\ell^\ell(x) = \frac{(-1)^\ell (1-x^2)^{\ell/2}}{2^\ell \ell!} \frac{d^{2\ell}}{dx^{2\ell}} \left[x^2 - 1 \right]^\ell$$

Now, $(x^2 - 1)^\ell = x^{2\ell} + \text{other terms}$. But only the highest power survives after 2ℓ derivatives. so we have,

$$\frac{d^{2\ell}}{dx^{2\ell}} \left[x^2 - 1 \right]^\ell = \frac{d^{2\ell}}{dx^{2\ell}} \left[x^{2\ell} \right] = 2\ell \frac{d^{2\ell-1}}{dx^{2\ell-1}} \left[x^{2\ell-1} \right] = (2\ell)(2\ell-1) \dots \frac{dx}{dx} = (2\ell)! \quad (1.110)$$

Putting it all together we have :

$$\begin{aligned} Y_\ell^\ell(\theta, \phi) &= \sqrt{\frac{2\ell+1}{4\pi}} \frac{1}{\sqrt{(2\ell)!}} e^{i\ell\phi} \times \frac{(-1)^\ell (1 - \cos^2 \theta)^{\ell/2}}{2^\ell \ell!} (2\ell)! \\ &= \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)(2\ell)!}{4\pi}} e^{i\ell\phi} (\sin^2 \theta)^{\ell/2} \\ &= \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} e^{i\ell\phi} (\sin \theta)^\ell \text{(QED)} \end{aligned}$$

1.9 Alternative way to get the eigenfunctions using “Ladder” operators.

Beforehand we had solved Eq. (1.76) to obtain the closed form of $f_\ell^m(\theta, \phi)$. We take the form of $f_\ell^m(\theta, \phi)$ as per Eq. (1.78) and using Eq. (1.80) we arrive at :

$$f_\ell^m(\theta, \phi) = \frac{e^{im\phi}}{\sqrt{2\pi}} \Theta_{\ell m}(\theta) \quad (1.111)$$

Taking $m = \ell$ and using Eq. (1.77) we get :

$$\hat{L}_+ f_\ell^\ell(\theta, \phi) = 0. \quad (1.112)$$

Suppose we only know this, and nothing else. We now wish to find out the form of $f_\ell^\ell(\theta, \phi)$. We can obtain this by solving the PDE using the form of $\hat{L}_+$ given in Eq. (1.74d) which we will do now.

$$\begin{aligned} \hat{L}_+ f_\ell^\ell(\theta, \phi) &= \hbar e^{i\phi} \left[\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right] \left[\frac{e^{i\ell\phi}}{\sqrt{2\pi}} \Theta_{\ell\ell}(\theta) \right] \\ &= \hbar e^{i\phi} \left[\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right] \left[\frac{e^{i\ell\phi}}{\sqrt{2\pi}} \Theta_{\ell\ell}(\theta) \right] \\ &= \frac{\hbar}{\sqrt{2\pi}} e^{(\ell+1)\phi} \left[\frac{d\Theta_{\ell\ell}(\theta)}{d\theta} + i^2 \ell \cot \theta \Theta_{\ell\ell}(\theta) \right] = 0 \end{aligned}$$

Which gives :

$$\begin{aligned} \frac{d\Theta_{\ell\ell}(\theta)}{d\theta} &= \ell \cot \theta \Theta_{\ell\ell}(\theta) \\ \frac{d\Theta_{\ell\ell}(\theta)}{\Theta_{\ell\ell}(\theta)} &= \ell \cot \theta d\theta = \ell \frac{d \sin \theta}{\sin \theta} \\ \ln[\Theta_{\ell\ell}(\theta)] &= \ell \ln(\sin \theta) + \ln A \end{aligned}$$

Which gives the solution for $\Theta_{\ell\ell}(\theta)$ as :

$$\Theta_{\ell\ell}(\theta) = A (\sin \theta)^\ell \quad (1.113)$$

All we need to find is the constant A so that $f_\ell^\ell(\theta, \phi)$ is normalized. The form of $f_\ell^\ell(\theta, \phi)$ is given by :

$$f_\ell^\ell(\theta, \phi) = \frac{A}{\sqrt{2\pi}} e^{i\ell\phi} (\sin \theta)^\ell \quad (1.114)$$

Using Eq. (1.96) we have :

$$\int d\Omega [f_\ell^\ell(\theta, \phi)]^* f_\ell^\ell(\theta, \phi) = 1$$

After performing the ϕ equation we are left with the condition that :

$$|A|^2 \int_0^\pi \sin \theta d\theta (\sin \theta)^{2\ell} = 1$$

The integral is done in the chapter B. Using the results we get the value of A as :

$$A = \frac{1}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{2}} \quad (1.115)$$

Finally we obtain the form of $f_\ell^\ell(\theta, \phi)$:

$$f_\ell^\ell(\theta, \phi) = \frac{1}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} \left[e^{i\phi} \sin \theta \right]^\ell \quad (1.116)$$

Comparing Eq. (1.108) and Eq. (1.116) we see that the results are exactly the same but we are missing a factor of $(-1)^\ell$, which is arbitrary any ways.²⁵ Let us pause for a moment and appreciate what we have achieved here. Without going to solve for the generic Legendre differential equation, we have used certain facts and cleverly obtained the solution. Now if we want to find out the other states, all we need to do is nothing but to apply $\hat{L}_-$ on this state $f_\ell^\ell(\theta, \phi)$ again and again. By the repeated action of $\hat{L}_-$ we can get all the states and we have to use Eq. (1.77) to obtain normalized solutions so that every time we do the operation we don't have to normalize again and again.

To illustrate further, we do one more step by calculating the action of $\hat{L}_-$ on the state $f_\ell^\ell(\theta, \phi)$. Using the form of $\hat{L}_-$ as in Eq. (1.74d) and for normalization using Eq. (1.77) we get :

$$\begin{aligned}\hat{L}_- f_\ell^\ell(\theta, \phi) &= \hbar \sqrt{\ell(\ell+1) - \ell(\ell-1)} f_\ell^{\ell-1}(\theta, \phi) = \hbar \sqrt{2\ell} f_\ell^{\ell-1}(\theta, \phi) \\ &= -\hbar e^{-i\phi} \left[\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \phi} \right] f_\ell^\ell(\theta, \phi) \\ &= -\hbar e^{-i\phi} \left[\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \phi} \right] \left[\frac{1}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} (e^{i\phi} \sin \theta)^\ell \right] \\ &= -\frac{\hbar}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} e^{i(\ell-1)\phi} \left[\frac{d}{d\theta} (\sin \theta)^\ell + \ell \cot \theta (\sin \theta)^\ell \right]\end{aligned}$$

Which gives :

$$f_\ell^{\ell-1}(\theta, \phi) = \left[\frac{1}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} \times \frac{1}{2\ell} \right] e^{i(\ell-1)\phi} \left[\frac{d}{d\theta} (\sin \theta)^\ell + \ell \cot \theta (\sin \theta)^\ell \right]^{26} \quad (1.117)$$

And this process goes on and on. But at some point we will hit the bottom rung of the ladder i.e. $f_\ell^{-\ell}(\theta, \phi)$.

Problem 1.30 Finding the form of $Y_\ell^{-\ell}(\theta, \phi)$

Write down the form of $Y_\ell^{-\ell}(\theta, \phi)$ without doing any calculations.

Hint : Use Eq. (1.100) and **Example (1.12)**.

Problem 1.31 Ascending the “Ladder”

(a). Following the same procedure as we did in Sec. 1.9 find out the form of $f_\ell^{-\ell}(\theta, \phi)$ up to a normalization constant.

Hint : Take the form of $\hat{L}_-$ as in Eq. (1.74d) and the form of $f_\ell^m(\theta, \phi)$ as given by Eq. (1.111). Also use the fact that :

$$\hat{L}_- f_\ell^{-\ell}(\theta, \phi) = 0. \quad (1.118)$$

(b). Using some expressions found out in Sec. 1.9 determine the normalization constant.

(c). Compare the form of $f_\ell^{-\ell}(\theta, \phi)$ with the form of $Y_\ell^{-\ell}(\theta, \phi)$ obtained in **Problem (1.30)**. Do you find any discrepancy.

Comment: You will find that the results are exactly matching, not even a factor off (the problem we faced in Sec. 1.9). So the best way to do this is you start with the bottom rung of the ladder i.e. $f_\ell^{-\ell}(\theta, \phi)$ and by the repeated action of $\hat{L}_+$ ascend the ladder.

²⁵If you think a little harder, this factor is nothing but the $(-1)^m$ factor, Condon-Shortley phase which we introduced in that section.

²⁶Note that we still haven't included the $(-1)^\ell$ factor that will be carried from $f_\ell^\ell(\theta, \phi)$ to $f_\ell^{\ell-1}(\theta, \phi)$ and so on.

Problem 1.32

Following Eq. (1.77) we write :

$$\hat{L}_{\pm} Y_{\ell}^m(\theta, \phi) = \hbar \sqrt{\ell(\ell+1) - m(m \pm 1)} Y_{\ell}^{m \pm 1}(\theta, \phi) \quad (1.119)$$

Eq. (1.119) is a very useful relation as it allows us to move through the ladder. For instance if we know one $Y_{\ell}^m(\theta, \phi)$ (any one of it), by the application of $\hat{L}_{\pm}$ we can very easily find out the remaining ones. Now given :

$$Y_3^0(\theta, \phi) = \sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta) \quad (1.120)$$

- (a) Find out $Y_3^{\pm 1}(\theta, \phi)$ by using ladder operators on $Y_3^0(\theta, \phi)$. For instance, using Eq. (1.119) we can write $\hat{L}_{\pm} Y_3^0(\theta, \phi) = \hbar \sqrt{12} Y_3^{\pm 1}(\theta, \phi)$. Using Eq. (1.74d) find out the states explicitly. You should get :

$$Y_3^{\pm 1}(\theta, \phi) = \mp \left(\frac{21}{64\pi} \right)^{1/2} \sin \theta (5 \cos^2 \theta - 1), e^{\pm i\phi} \quad (1.121)$$

- (b) Write $Y_3^{\pm 1}(\theta, \phi)$ in terms of (x, y, z) .
- (c) Using Eq. (1.7c) find out $\hat{L}_z Y_3^{\pm 1}(\theta, \phi)$.
- (d) Using the form of $\hat{\mathbf{L}}^2$ in spherical polar coordinates given by Eq. (1.23) find out $\hat{\mathbf{L}}^2 Y_3^{\pm 1}(\theta, \phi)$.
- (e) You can also try to do it in Cartesian coordinates. Do it if you can.
- (f) Can you check $Y_3^{\pm 1}(\theta, \phi)$ are individually normalized and they are orthogonal to each other by doing the integration explicitly.

Note that once we get $Y_3^1(\theta, \phi)$ we can directly write $Y_3^{-1}(\theta, \phi)$ using Eq. (1.100).

Problem 1.33

In **Problem (1.26)** we tabulated the Spherical Harmonics for $\ell = 2$. Start with the state for which $\ell = 2$ and $m = -2$. You can obtain the state $Y_2^{-2}(\theta, \phi)$ using Eq. (1.100) and Eq. (1.108). Now by the repeated action of $\hat{L}_{+}$ on this state obtain the all the other states. Don't forget to normalize at each step. Use Eq. (1.119) so that you don't have to normalize at each step. Compare the results tabulated in **Problem (1.26)**.

Comments :

Start with generic form of $\Theta_{\ell m}(\theta) = A(\sin \theta)^{\ell}$. Putting $\ell = 2$, normalize it to obtain A . Voilà! You get $Y_2^{-2}(\theta, \phi)$.

1.10 Few Applications

1.10.1 Energy Levels of a “Rigid Rotator”

This is the same problem for a particle of mass M which is constrained to move on a surface of sphere of radius $r = R$ but free otherwise. Classically the Lagrangian $\mathcal{L}$ is given by :

$$\mathcal{L}(\dot{\theta}, \dot{\phi}, \theta) = \frac{MR^2}{2} \left[\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right] \quad (1.122)$$

The canonical momentum is defined as :

$$P_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = MR^2 \dot{\theta} \quad (1.123a)$$

$$P_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = MR^2 \sin^2 \theta \dot{\phi} \quad (1.123b)$$

Using them we get the “Hamiltonian” as :

$$\mathcal{H} = P_\theta \dot{\theta} + P_\phi \dot{\phi} - \mathcal{L} = \frac{1}{2MR^2} \left[P_\theta^2 + \frac{P_\phi^2}{\sin^2 \theta} \right] = \frac{1}{2I} \left[P_\theta^2 + \frac{P_\phi^2}{\sin^2 \theta} \right]^{27}$$

Now in terms of angular momentum $\vec{\mathbf{L}}$ we have $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = MR^2 \left[\hat{\mathbf{e}}_\theta \dot{\theta} + \hat{\mathbf{e}}_\phi \sin \theta \dot{\phi} \right]$. We now evaluate the magnitude of $\vec{\mathbf{L}}$, i.e.

$$\begin{aligned} |\vec{\mathbf{L}}|^2 &= M^2 R^4 \left[\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right] \\ &= [MR^2 \dot{\theta}]^2 + \frac{[MR^2 \sin^2 \theta \dot{\phi}]^2}{\sin^2 \theta} \\ &= P_\theta^2 + \frac{P_\phi^2}{\sin^2 \theta} \end{aligned}$$

Which finally gives :

$$\mathcal{H} = \frac{|\vec{\mathbf{L}}|^2}{2I} \quad (1.124)$$

Classically all the “positive energy values” are allowed, as there is no restriction on what $|\vec{\mathbf{L}}|$ can take. In order to analyse the same case quantum mechanically, what we have to do is to write the Hamiltonian in operator form and then proceed to solve the so called “Sacred Schrödinger Equation”. We replace the observables by their corresponding operators as given by Eq. (1.124).²⁸

$$\hat{H} = \frac{\hat{\mathbf{L}}^2}{2I} \quad (1.125)$$

The corresponding “Schrödinger Equation” is given by :

$$\hat{H} \psi(\theta, \phi) = E \psi(\theta, \phi) \Rightarrow \hat{\mathbf{L}}^2 \psi(\theta, \phi) = 2EI \psi(\theta, \phi) \quad (1.126)$$

We already know that spherical harmonics are the eigenfunctions of $\hat{\mathbf{L}}^2$ and eigenvalues are quantized. So we have the solution of wave-function $\psi(\theta, \phi) = Y_\ell^m(\theta, \phi)$, with $2EI = \ell(\ell + 1)\hbar^2$, which gives the discrete energy eigenvalues :

$$E_\ell = \frac{\hbar^2}{2I} \ell(\ell + 1), \quad \text{where } \ell = 0, 1, 2, \dots \quad (1.127)$$

So for a given ℓ there are $(2\ell + 1)$ no of states with the same energy eigenvalue E_ℓ . So each energy level is $(2\ell + 1)$ fold degenerate and corresponding eigenfunctions are $Y_\ell^\ell(\theta, \phi), Y_\ell^{\ell-1}(\theta, \phi), \dots, Y_\ell^{-\ell+1}(\theta, \phi), Y_\ell^{-\ell}(\theta, \phi)$.

²⁷Where we have written $I = MR^2$

²⁸ $\mathcal{H} \rightarrow \hat{H}, |\vec{\mathbf{L}}|^2 \rightarrow \hat{\mathbf{L}}^2$

You might ask why “what is the reason behind this degeneracy”. The answer is very simple. We can see the Hamiltonian is independent of the orientation of $\vec{L}$ in space. So it does not depend on which direction $\vec{L}$ is pointing at. “Classically” all directions are allowed, but in QM due to quantized nature of $\hat{L}^2$ we have only $(2\ell + 1)$ no of discrete orientations possible.

1.10.2 Rotational Energy Levels of Diatomic Molecule

So far we have discussed about the *Rigid Rotator*. But you might wanna ask why we have studied that. There must be some reason behind this, right ?? Before connecting it with a real life problem let us discuss a bit about the two body problem.

Consider two particles of mass m_1 and m_2 respectively and the distance is $\vec{r}$ which is say fixed. Now this is a classic “two body” problem which can be reduced to one body problem with the concept of “reduced mass”. We achieve this by going to COM frame from any arbitrary frame. We can think the same problem as in COM frame we have a particle of mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ at a distance $\vec{r}$ which is shown in Fig (1.5).

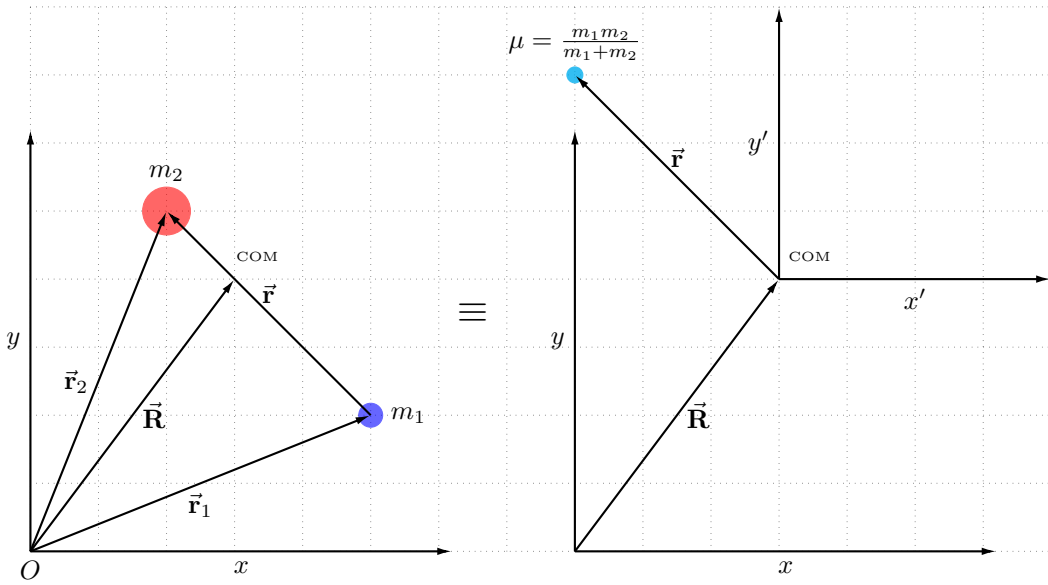


Figure 1.5: Two body problem diagram with mass $m_1 = 1$ unit and $m_2 = 2$ unit. The position vectors are $\vec{r}_1 = (5, 2, 0)$ and $\vec{r}_2 = (2, 5, 0)$, which gives the coordinates of COM frame $\vec{R} = (3, 4, 0)$. The relative position vector is given by $\vec{r} = \vec{r}_2 - \vec{r}_1 = (-3, 3, 0)$. The reduced mass here becomes $\mu = \frac{2}{3}$ units. The masses are filled with a circle whose radius is proportional to the mass. For ease of visibility “one unit mass” equal to radius of “5pt”.

Now coming to application. Consider a diatomic molecule with a bond length a_0 which is fixed. Now what we want to know is that “quantum mechanically what is the energy spectrum”. Naively to solve this we have to write down the Hamiltonian of this two particle system and then write down the corresponding operator form. Afterwards by solving “Schrödinger Equation” we can get the energy spectrum. Now this is a bit complicated as this a constrained system. The separation between two of them remains constant whereas there is no restriction on $\vec{r}_1$ or $\vec{r}_2$ separately. What we do is that we convert this problem to a one body problem (in COM frame) which is constrained to move on a sphere of radius $|\vec{r}| = a_0$. Then can write down the Hamiltonian of the system as we did in case of “Rigid Rotator”. Following Eq. (1.127) we write the energy levels of this “diatomic molecule” as :

$$E_J = \frac{\hbar^2}{2I} J(J+1),^{29} \quad \text{where } J = 0, 1, 2, \dots$$

Where $I = \mu a_0^2$ and by writing $D = \frac{\hbar^2}{2I} = \frac{\hbar^2}{2\mu a_0^2}$ we get the energy spectrum :

$$E_J = DJ(J+1) \tag{1.128}$$

²⁹Instead of ℓ we write J here, nothing serious.

So we see the energy spectrum is discrete. Now consider the following scenario ; we put this molecule in an external radiation field. The molecule will absorb those frequencies of the radiation which will be exactly same as the difference between it's two energy levels. The allowed transitions are given by $\Delta J = \pm 1$. This is diagrammatically shown in Fig (1.6).

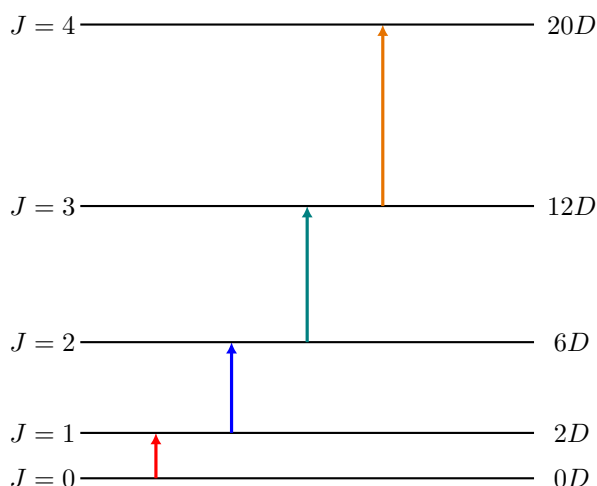


Figure 1.6: Energy levels of “diatomic molecule”. The arrow shows the allowed transitions.

Now consider transition from a state J to $J + 1$, for which the energy difference is $\Delta E_{J \rightarrow J+1} = D(J+1)(J+2) - DJ(J+1) = 2D(J+1)$. So the molecule will absorb the radiation having frequency $\nu_{J \rightarrow J+1} = \Delta E_{J \rightarrow J+1}/h = 2B(J+1)$.³⁰ Note that the separation between two adjacent frequency is $2B$. Basically we will see some dark spots in the absorption spectra of the molecule with fixed spacing $2B$. See Fig (1.7). Those are nothing but the “absorption lines” of the molecule and just looking at these we can know a lot about the molecule itself. See **Problem (1.34)** to get a taste of this.

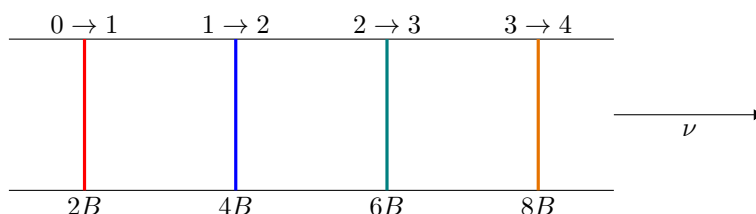


Figure 1.7: The absorption lines with fixed separation $2B$.

Problem 1.34

Consider the molecule “CO” made out of $^{12}_6\text{C}$ and $^{16}_8\text{O}$. The frequency separation $2B$ is obtained by studying the “absorption spectra” of the molecule. Now using the masses $m_{^{12}\text{C}} = 12 \text{ u}$ and $m_{^{16}\text{O}} = 15.9949146 \text{ u}$ and also $2B = 3.84235 \text{ cm}^{-1}$ find out the bond length a_0 in n.m. units. Where $1 \text{ u} = 1.6605390666 \times 10^{-27} \text{ kg}$. In order to convert from cm^{-1} unit to frequency unit you have to multiply it by the speed of light $c = 3 \times 10^{10} \text{ cm/s}$.

³⁰Here $B = D/h = \frac{h}{8\pi^2 I}$

1.10.3 Energy Levels of a Symmetric Top Molecule

Classically the Hamiltonian of a asymmetric top is given by :

$$\mathcal{H} = \frac{L_x^2}{2I_1} + \frac{L_y^2}{2I_2} + \frac{L_z^2}{2I_3} \quad (1.129)$$

Consider a symmetric top molecule with $I_1 = I_2 \neq I_3$ ³². Following Eq. (1.129) the corresponding quantum mechanical Hamiltonian operator is given by :

$$\hat{H} = \frac{\hat{L}_x^2}{2I_1} + \frac{\hat{L}_y^2}{2I_2} + \frac{\hat{L}_z^2}{2I_3} \quad (1.130)$$

Now with $I_1 = I_2$ we write Eq. (1.130) as :

$$\begin{aligned} \hat{H} &= \frac{\hat{L}_x^2 + \hat{L}_y^2}{2I_1} + \frac{\hat{L}_z^2}{2I_3} \\ &= \frac{\hat{\mathbf{L}}^2}{2I_1} + \left(\frac{1}{2I_3} - \frac{1}{2I_1} \right) \hat{L}_z^2 \end{aligned}$$

As the Hamiltonian commutes with $\hat{\mathbf{L}}^2$, they can have simultaneous eigenstates. Following the same procedure as we did in case of “rigid rotator” we immediately get the eigenstates $\psi(\theta, \phi) = Y_\ell^m(\theta, \phi)$ with energy eigenvalue :

$$E_{\ell, m} = \frac{\ell(\ell+1)\hbar^2}{2I_1} + \left(\frac{1}{2I_3} - \frac{1}{2I_1} \right) m^2 \hbar^2 = B\ell(\ell+1) + (C-B)m^2 \quad (1.131)$$

Looking at Eq. (1.131) we see that for a given ℓ there are states which are 2 fold degenerate corresponds to $\pm m$ for $m \neq 0$ with states $Y_\ell^{\pm m}(\theta, \phi)$. But non-degenerate for $m = 0$. Now one thing to notice is that if we put $I_1 = I_2 = I_3$ which gives $B = C$, we get back the rigid rotator eigenvalue. You might be thinking from where this “2 fold degeneracy comes”. The answer is the Hamiltonian is invariant under the exchange of axes $x \leftrightarrow y$.

In Eq. (1.131) if we put $I_1 = I_2 = I_3 = I$ (say), then the second term cancels and we get back the same energy eigenvalue solution what we had obtained in Eq. (1.127).

Based on the discussion above, two possibilities can occur: either

$$I_1 = I_2 > I_3 \quad \text{or} \quad I_1 = I_2 < I_3$$

For example, the molecule NH_3 falls into the first category, for which

$$I_1 = I_2 > I_3 \Rightarrow B > C$$

The shape of the ammonia molecule is *prolate*. On the other hand, for the molecule BH_3 ,

$$I_1 = I_2 < I_3 \Rightarrow B < C$$

and the shape of the BH_3 molecule is *oblate*. Thus, depending on the sign of $C - B$, the rotational energy spectrum changes, and consequently the characteristic absorption line features also change. By analysing these absorption spectra, we can infer the *shape of the molecule* upto some extent.

³¹Where I_1, I_2, I_3 are moment of inertia along different axes.

³²As $I_1 = I_2$ we are calling it symmetric top. If all are same then this will be the same as “rigid rotator”.

³³Where we have written $\frac{\hbar^2}{2I_1} = B$ and $\frac{\hbar^2}{2I_3} = C$.

1.11 Stern- Gerlach Experiment

1.12 Spin Angular Momentum

The theory of spin is identical to the general theory of angular momentum. By analogy with vector momentum, Spin is also represented by vector $\hat{\mathbf{S}}$ with components $\hat{S}_x, \hat{S}_y, \hat{S}_z$, which satisfy the fundamental commutation algebra :

$$[\hat{S}_i, \hat{S}_j] = i\hbar \epsilon_{ijk} \hat{S}_k \quad (1.132)$$

We get eigen-kets by $|s, m_s\rangle$. The eigenvalue equations are given as :

$$\hat{\mathbf{S}}^2 |s, m_s\rangle = s(s+1)\hbar^2 |s, m_s\rangle \quad (1.133a)$$

$$\hat{S}_z |s, m_s\rangle = m_s \hbar |s, m_s\rangle \quad (1.133b)$$

Where $m_s = -s, -s+1, \dots, s-1, s$. The ladder operators are given by :

$$\hat{S}_{\pm} = \hat{S}_x \pm i\hat{S}_y \quad (1.134)$$

The action of $\hat{S}_{\pm}$ on the state $|s, m_s\rangle$ is given by :

$$\hat{S}_{\pm} |s, m_s\rangle = \hbar \sqrt{s(s+1) - m_s(m_s \pm 1)} |s, m_s \pm 1\rangle \quad (1.135)$$

Also we have :

$$\langle \hat{S}_x \rangle = \langle \hat{S}_y \rangle = 0 \quad (1.136a)$$

$$\langle \hat{S}_x^2 \rangle = \langle \hat{S}_y^2 \rangle = \frac{\hbar^2}{2} [s(s+1) - m_s^2] \quad (1.136b)$$

They are also orthonormal and forms a complete basis. By this we mean that :

$$\langle s', m_{s'} | s, m_s \rangle = \delta_{s,s'} \delta_{m_s, m_{s'}} \quad (1.137a)$$

$$\sum_{m_s=-s}^s |s, m_s\rangle \langle s, m_s| = \hat{I} \quad (1.137b)$$

This is all you need to know about spin angular momentum. Unlike the orbital angular momentum case, there is no reason to exclude “half integers”. So s can take $0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ any value.

Every elementary particle has a specific value of s , which we name “spin”. For example π mesons have spin “0”. Electrons (e^-), protons (p), neutrons (n) and quarks have spin “ $\frac{1}{2}$ ”. Photon (γ) has spin “1”. The delta particles Δ have spin $\frac{3}{2}$.

Integer spin particles are called bosons (pions, mesons, photons, gravitons) and half integer particles are called “fermions” (electrons, quarks, etc).

1.12.1 Spin $\frac{1}{2}$ particle and Pauli matrices

For a spin half particle, $s = \frac{1}{2}$. So m_s can take two values, either $m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$. Thus the particle is either in $|s, m_s\rangle = |\frac{1}{2}, +\frac{1}{2}\rangle$ state or $|s, m_s\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$ state or in a superposition state of them.³⁵ The previous one we call “spin up”, informally denoted by $|\uparrow\rangle$, while the latter one is called “spin down” denoted by $|\downarrow\rangle$. The eigenvalue equations

³⁴ $\hat{I}$ here is a $(2s+1) \times (2s+1)$ identity matrix.

³⁵ See Eq. (1.150).

Eq. (1.133) are now written as :

$$\hat{\mathbf{S}}^2 \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle = \frac{3\hbar^2}{4} \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle \quad (1.138a)$$

$$\hat{S}_z \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle = \pm \frac{\hbar}{2} \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle \quad (1.138b)$$

As shown in Sec. 1.7, we can represent this spin graphically by a vector of length $|\vec{\mathbf{S}}| = \sqrt{\frac{1}{2}(\frac{1}{2} + 1)}\hbar = \frac{\sqrt{3}}{2}\hbar$ rotating along the surface of a cone with half angle :

$$\theta_{\pm} = \arccos\left(\frac{\pm \frac{1}{2}}{\sqrt{\frac{1}{2}(\frac{1}{2} + 1)}}\right) = \arccos\left(\pm \frac{1}{\sqrt{3}}\right) = \pm 54.7356^\circ \quad (1.139)$$

Only two configurations are possible. The projection of $\hat{\mathbf{S}}$ on the ‘ z ’ axis can take only two values, i.e. $\pm \frac{\hbar}{2}$. Now the matrix representation of $\hat{\mathbf{S}}^2$ is given as :

$$\begin{aligned} \hat{\mathbf{S}}^2 &= \begin{pmatrix} \langle \frac{1}{2}, +\frac{1}{2} | \hat{\mathbf{S}}^2 | \frac{1}{2}, +\frac{1}{2} \rangle & \langle \frac{1}{2}, +\frac{1}{2} | \hat{\mathbf{S}}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \\ \langle \frac{1}{2}, -\frac{1}{2} | \hat{\mathbf{S}}^2 | \frac{1}{2}, +\frac{1}{2} \rangle & \langle \frac{1}{2}, -\frac{1}{2} | \hat{\mathbf{S}}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \end{pmatrix} \\ &= \begin{pmatrix} \langle \uparrow | \hat{\mathbf{S}}^2 | \uparrow \rangle & \langle \uparrow | \hat{\mathbf{S}}^2 | \downarrow \rangle \\ \langle \downarrow | \hat{\mathbf{S}}^2 | \uparrow \rangle & \langle \downarrow | \hat{\mathbf{S}}^2 | \downarrow \rangle \end{pmatrix} \\ &= \frac{3\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned} \quad ^{36}$$

Following the same way we get the representation of $\hat{S}_z$ as follows :

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.140)$$

And also using Eq. (1.135) we have :

$$\hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \hat{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad (1.141)$$

Now using Eq. (1.134) we have,

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (1.142)$$

Now from eigenvalue equation Eq. (1.138b) we have :

$$\begin{aligned} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix} &= \pm \frac{\hbar}{2} \begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix} \\ \Rightarrow \begin{pmatrix} +a_{\pm} \\ -b_{\pm} \end{pmatrix} &= \begin{pmatrix} \pm a_{\pm} \\ \pm b_{\pm} \end{pmatrix} \end{aligned} \quad ^{37}$$

Taking the ‘upper’ sign we get :

$$\Rightarrow \begin{cases} a_+ = a_+ \\ b_+ = -b_+ \end{cases} \quad (1.143)$$

³⁶For any operator $\hat{A}$ the matrix representation of that operator for spin “ $\frac{1}{2}$ ” particle is given as : $\hat{A} = \begin{pmatrix} \langle \uparrow | \hat{A} | \uparrow \rangle & \langle \uparrow | \hat{A} | \downarrow \rangle \\ \langle \downarrow | \hat{A} | \uparrow \rangle & \langle \downarrow | \hat{A} | \downarrow \rangle \end{pmatrix}$

³⁷Where we have represented the states $|\frac{1}{2}, \pm \frac{1}{2}\rangle$ as column vectors $\begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix}$.

gives the normalized solution for $|\frac{1}{2}, +\frac{1}{2}\rangle$ as

$$|\frac{1}{2}, +\frac{1}{2}\rangle = \begin{pmatrix} a_+ \\ b_+ \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.144)$$

And similarly we have the solution for the $|\frac{1}{2}, -\frac{1}{2}\rangle$ as

$$|\frac{1}{2}, -\frac{1}{2}\rangle = \begin{pmatrix} a_- \\ b_- \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.145)$$

We can easily verify that they are orthonormal,

$$\langle \uparrow | \uparrow \rangle = \langle \downarrow | \downarrow \rangle = 1 \quad (1.146)$$

and,

$$\langle \uparrow | \downarrow \rangle = \langle \downarrow | \uparrow \rangle = 0 \quad (1.147)$$

They form a complete basis as :

$$\begin{aligned} \sum_{m_s = -\frac{1}{2}}^{\frac{1}{2}} \left| \frac{1}{2}, m_s \right\rangle \left\langle \frac{1}{2}, m_s \right| &= \left| \frac{1}{2}, +\frac{1}{2} \right\rangle \left\langle \frac{1}{2}, +\frac{1}{2} \right| + \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \left\langle \frac{1}{2}, -\frac{1}{2} \right| \\ &= |\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow| \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{I}_{2 \times 2} \end{aligned}$$

Problem 1.35 Eigenvalue and eigenvectors of $\hat{S}_x$ and $\hat{S}_y$

Find out the eigenvalues and eigenstates of $\hat{S}_x$ and $\hat{S}_y$ and write them in terms of $|\uparrow\rangle$ and $|\downarrow\rangle$ kets.

Hint : Use the matrix form of those operators $\hat{S}_x, \hat{S}_y$ and also the column vector form of the basis states $|\uparrow\rangle, |\downarrow\rangle$.

$$|\psi_x\rangle_{\pm} = \frac{1}{\sqrt{2}} \left[|\uparrow\rangle \pm |\downarrow\rangle \right] = \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} \quad (1.148a)$$

$$|\psi_y\rangle_{\pm} = \frac{1}{\sqrt{2}} \left[|\uparrow\rangle \pm i |\downarrow\rangle \right] = \begin{pmatrix} 1 \\ \pm i \end{pmatrix} \quad (1.148b)$$

Where :

$$\hat{S}_x |\psi_x\rangle_{\pm} = \pm \frac{\hbar}{2} |\psi_x\rangle_{\pm} \quad (1.149a)$$

$$\hat{S}_y |\psi_y\rangle_{\pm} = \pm \frac{\hbar}{2} |\psi_y\rangle_{\pm} \quad (1.149b)$$

Using the basis states $|\uparrow\rangle, |\downarrow\rangle$, we can write the most generic state describing a spin “ $\frac{1}{2}$ ” particle as :

$$|\psi\rangle = a |\uparrow\rangle + b |\downarrow\rangle \quad (1.150)$$

Using the fact the state is normalized, i.e. $\langle\psi|\psi\rangle = 1$, we infer that :

$$|a|^2 + |b|^2 = 1 \quad (1.151)$$

Now for this state if we measure the ‘ z ’ component of the spin what is $\langle\hat{S}_z\rangle$, then the probability of finding the particle in “up” state $|\uparrow\rangle$ is $|\langle\uparrow|\psi\rangle|^2 = |a|^2$ and similarly in the “down” state $|\downarrow\rangle$ is $|\langle\downarrow|\psi\rangle|^2 = |b|^2$.³⁸

Problem 1.36

(a). Consider the generic state as given in Eq. (1.150). Now if we measure the ‘ x ’ component of the spin, then what are the probability of getting $\langle\hat{S}_x\rangle = \frac{\hbar}{2}$ and $-\frac{\hbar}{2}$ respectively.

Hint : First write down generic state $|\psi\rangle$ in terms of eigenstates of the operator $\hat{S}_x$, i.e. $|\psi_x\rangle_{\pm}$. Now using $|\psi_x\rangle_{\pm} = \frac{1}{\sqrt{2}} [|\uparrow\rangle \pm |\downarrow\rangle]$, we write :

$$|\uparrow\rangle = \frac{1}{2} [|\psi_x\rangle_+ + |\psi_x\rangle_-] \quad (1.152a)$$

$$|\downarrow\rangle = \frac{1}{2} [|\psi_x\rangle_+ - |\psi_x\rangle_-] \quad (1.152b)$$

Which gives :

$$|\psi\rangle = \frac{a+b}{\sqrt{2}} |\psi_x\rangle_+ + \frac{a-b}{\sqrt{2}} |\psi_x\rangle_- \quad (1.153)$$

Remember for any operator $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$.

(b). Now what are the probabilities of getting ‘ y ’ component, $\langle\hat{S}_y\rangle = \pm\frac{\hbar}{2}$ respectively.

Proceed in the same way as done in the above part.

$$|\psi\rangle = \frac{a-ib}{\sqrt{2}} |\psi_y\rangle_+ + \frac{a+ib}{\sqrt{2}} |\psi_y\rangle_- \quad (1.154)$$

Problem 1.37 Action of $\hat{S}_x$ and $\hat{S}_y$ on the basis state $|\uparrow\rangle$ and $|\downarrow\rangle$

Write down the action of $\hat{S}_x, \hat{S}_y$ on the states $|\uparrow\rangle, |\downarrow\rangle$, i.e. to find :

(a) $\hat{S}_x |\uparrow\rangle$

(b) $\hat{S}_x |\downarrow\rangle$

(c) $\hat{S}_y |\uparrow\rangle$

(d) $\hat{S}_y |\downarrow\rangle$

Hint: Use the matrix representation of the operators and also the column vector form of the basis states $|\uparrow\rangle, |\downarrow\rangle$. Don’t forget to express them in terms of $|\uparrow\rangle, |\downarrow\rangle$.

Problem 1.38

Consider a normalized state :

$$|\chi\rangle = A \begin{pmatrix} 3i \\ 4 \end{pmatrix} \quad (1.155)$$

(a) Find the normalization constant A .

³⁸Although I said it that way, more precisely I should say, if we measure the ‘ z ’ component of spin then $|a|^2$ is the probability that we will get the ‘ z ’ component as $+\frac{\hbar}{2}$ and similarly $|b|^2$ is the probability of getting ‘ z ’ component as $-\frac{\hbar}{2}$.

(b) Find $\langle \hat{S}_x \rangle$, $\langle \hat{S}_y \rangle$ and $\langle \hat{S}_z \rangle$.

(c) Find the corresponding uncertainties, i.e. σ_{S_x} , σ_{S_y} and σ_{S_z} . Where $\sigma_{S_i} = \sqrt{\langle \hat{S}_i^2 \rangle - \langle \hat{S}_i \rangle^2}$.

(d) Are they consistent with the generalized uncertainty principle :

$$\sigma_{S_i} \sigma_{S_j} \geq \frac{\hbar}{2} \left| \langle \hat{S}_k \rangle \right| \epsilon_{ijk} \quad (1.156)$$

Here take all the ‘ i, j ’ pairs (xy, yz, zx) .

Now $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ all carry a factor of $\frac{\hbar}{2}$. It is nice to write it in the form :

$$\hat{\mathbf{S}} = \frac{\hbar}{2} \vec{\sigma} \quad (1.157)$$

Where $\vec{\sigma} = \sigma_x \hat{\mathbf{i}} + \sigma_y \hat{\mathbf{j}} + \sigma_z \hat{\mathbf{k}}$. The matrix representation is given by :

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.158)$$

They are famously known as the “Pauli matrices”, named after the physicist “Wolfgang Pauli”. One important thing is to notice that all pauli matrices are hermitian ³⁹, traceless and having determinant -1. i.e.

$$\sigma_i = \sigma_i^\dagger, \quad \text{Tr}(\sigma_i) = 0, \quad \det(\sigma_i) = -1 \quad (1.159)$$

Problem 1.39 Anti-Commutator of Pauli Matrices

Prove that :

$$\{\sigma_j, \sigma_k\} = 2\hat{\mathbb{I}}_{2 \times 2} \delta_{jk} \quad (1.160)$$

Where $j, k = x, y, z$.

Problem 1.40 Product rule of Pauli Matrices

Prove that Pauli matrices satisfy the following product rule :

$$\sigma_j \sigma_k = \delta_{jk} \hat{\mathbb{I}}_{2 \times 2} + i \epsilon_{jkl} \sigma_l \quad (1.161)$$

Note that in the second term, it is summed over l .

Hint : Using Eq. (1.132) and Eq. (1.157) we can write :

$$[\sigma_j, \sigma_k] = 2i \epsilon_{jkl} \sigma_l \quad (1.162)$$

Adding Eq. (1.160) and Eq. (1.162) will give you the required answer.

Example 1.13 Dot product with Pauli Matrices

Prove that :

$$(\vec{\sigma} \cdot \vec{a})(\vec{\sigma} \cdot \vec{b}) = (\vec{a} \cdot \vec{b}) \hat{\mathbb{I}}_{2 \times 2} + i \vec{\sigma} \cdot (\vec{a} \times \vec{b}) \quad (1.163)$$

³⁹Of course, they represent physical observables, and operators corresponding to physical observables must be hermitian.

Ans :

$$\begin{aligned}
\text{L.H.S.} &= (\vec{\sigma} \cdot \vec{a})(\vec{\sigma} \cdot \vec{b}) \\
&= (\sigma_i a_i)(\sigma_j b_j) \\
&= \sigma_i \sigma_j (a_i b_j) \\
&= [\delta_{ij} \hat{I}_{2 \times 2} + i \epsilon_{ijk} \sigma_k] (a_i b_j) \\
&= a_i b_i \hat{I}_{2 \times 2} + i \sigma_k (\epsilon_{ijk} a_i b_j) \\
&= \vec{a} \cdot \vec{b} \hat{I}_{2 \times 2} + i \sigma_k (\vec{a} \times \vec{b})_k \\
&= (\vec{a} \cdot \vec{b}) \hat{I}_{2 \times 2} + i \vec{\sigma} \cdot (\vec{a} \times \vec{b}) = \text{R.H.S. (QED)}
\end{aligned}$$

Problem 1.41 Exponential of Pauli Matrices

Show that for a real number α ,

$$e^{i\alpha\sigma_j} = \hat{I}_{2 \times 2} \cos \alpha + i\sigma_j \sin \alpha \quad (1.164)$$

Hint: Use the expansion $e^{\mathbb{M}} = 1 + \mathbb{M} + \frac{\mathbb{M}^2}{2!} + \frac{\mathbb{M}^3}{3!} + \dots$. Where $\mathbb{M} = i\alpha\sigma_j$. Now separate out the odd and even powers of σ_j and use $\sigma_j^2 = \hat{I}_{2 \times 2}$.

$$\sigma_j^n = \begin{cases} \hat{I}_{2 \times 2} & \text{if } n \text{ is even.} \\ \sigma_j & \text{if } n \text{ is odd.} \end{cases} \quad (1.165)$$

Proof :

let $n = 2m$. Then $\sigma_j^{2m} = \underbrace{\sigma_j^2 \cdot \sigma_j^2 \cdots \sigma_j^2}_{n \text{ times}} = \hat{I}_{2 \times 2}$.

And for the odd case $n = 2m + 1$, $\sigma_j^{2m+1} = \sigma_j \times \sigma_j^{2m} = \sigma_j$

Example 1.14 Spin Matrix along arbitrary direction for “spin half particle”

Find the eigenvalues and eigenstate of spin operator in a direction $\hat{n}$ for a spin half particle, i.e. $s = \frac{1}{2}$.

Ans:

Using $\hat{n} = (\sin \theta \cos \phi) \hat{i} + (\sin \theta \sin \phi) \hat{j} + \cos \theta \hat{k}$ and the form of spin half matrices $\hat{S}_x, \hat{S}_y$ and $\hat{S}_z$ we write the generic spin operator as :

$$\begin{aligned}
\hat{n} \cdot \hat{\mathbf{S}} &= \sin \theta \cos \phi \hat{S}_x + \sin \theta \sin \phi \hat{S}_y + \cos \theta \hat{S}_z \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}
\end{aligned}$$

The eigenvalue equation is given by :

$$\hat{n} \cdot \hat{\mathbf{S}} |\lambda\rangle = \lambda |\lambda\rangle \quad (1.166)$$

Eigenvalues:

$$\begin{aligned}
& \begin{vmatrix} \frac{\hbar}{2} \cos \theta - \lambda & \frac{\hbar}{2} \sin \theta e^{-i\phi} \\ \frac{\hbar}{2} \sin \theta e^{i\phi} & -\frac{\hbar}{2} \cos \theta - \lambda \end{vmatrix} = 0 \\
& \Rightarrow -\left(\frac{\hbar}{2} \cos \theta - \lambda\right) \left(\frac{\hbar}{2} \cos \theta + \lambda\right) = \frac{\hbar^2}{4} \sin^2 \theta \\
& \Rightarrow \lambda^2 - \frac{\hbar^2}{4} \cos^2 \theta = \frac{\hbar^2}{4} \sin^2 \theta \\
& \Rightarrow \lambda^2 = \frac{\hbar^2}{4} \cos^2 \theta + \frac{\hbar^2}{4} \sin^2 \theta = \frac{\hbar^2}{4} \\
& \Rightarrow \lambda = \pm \frac{\hbar}{2}
\end{aligned}$$

Eigenvectors: Writing $|\lambda_{\pm}\rangle = \begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix}$ and by $\hat{\mathbf{n}} \cdot \hat{\mathbf{S}} |\lambda_{\pm}\rangle = \pm \frac{\hbar}{2} |\lambda_{\pm}\rangle$ we get :

$$\begin{aligned}
& \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix} = \pm \frac{\hbar}{2} \begin{pmatrix} a_{\pm} \\ b_{\pm} \end{pmatrix} \\
& \Rightarrow \begin{cases} \cos \theta a_{\pm} + \sin \theta e^{-i\phi} b_{\pm} = \pm a_{\pm} \\ \sin \theta e^{i\phi} a_{\pm} - \cos \theta b_{\pm} = \pm b_{\pm} \end{cases}
\end{aligned}$$

Solving them basically gives us the same result : $b_{\pm} = \frac{\pm(1 \mp \cos \theta)}{\sin \theta} e^{i\phi} a_{\pm} = \frac{\sin \theta}{\pm(1 \pm \cos \theta)} e^{i\phi} a_{\pm}$. So finally we have :

$$b_{\pm} = \pm [\tan \frac{\theta}{2}]^{\pm 1} e^{i\phi} a_{\pm} \quad (1.167)$$

Using this and the normalization condition $|a_{\pm}|^2 + |b_{\pm}|^2 = 1$, we get

$$|a_{\pm}|^2 = \frac{1}{1 + [\tan \frac{\theta}{2}]^{\pm 2}}$$

Now taking the “+” sign gives $|a_{+}|^2 = [\cos \frac{\theta}{2}]^2 \Rightarrow a_{+} = \cos \frac{\theta}{2}$ and $b_{+} = \tan \frac{\theta}{2} e^{i\phi} a_{+} = e^{i\phi} \sin \frac{\theta}{2}$. And very similarly “-” sign gives $a_{-} = \sin \frac{\theta}{2}$ and $b_{-} = -[\tan \frac{\theta}{2}]^{-1} e^{i\phi} a_{-} = -e^{i\phi} \cos \frac{\theta}{2}$. Then we have the eigenstates as :

$$|\lambda_{+}\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}, \quad |\lambda_{-}\rangle = \begin{pmatrix} \sin \frac{\theta}{2} \\ -e^{i\phi} \cos \frac{\theta}{2} \end{pmatrix} \quad (1.168)$$

It is very easy to see that they are “complete and orthonormal”, as :

$$\langle \lambda_{+} | \lambda_{-} \rangle = \begin{pmatrix} \cos \frac{\theta}{2} & e^{-i\phi} \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \sin \frac{\theta}{2} \\ -e^{i\phi} \cos \frac{\theta}{2} \end{pmatrix} = \cancel{\cos \frac{\theta}{2} \sin \frac{\theta}{2}} - \cancel{\sin \frac{\theta}{2} \cos \frac{\theta}{2}} = 0$$

And also :

$$\begin{aligned}
 |\lambda_+\rangle\langle\lambda_+| + |\lambda_-\rangle\langle\lambda_-| &= \begin{pmatrix} \cos \frac{\theta}{2} & e^{-i\phi} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} + \begin{pmatrix} \sin \frac{\theta}{2} & -e^{-i\phi} \cos \frac{\theta}{2} \\ -e^{i\phi} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{pmatrix} \\
 &= \begin{pmatrix} \cos^2 \frac{\theta}{2} & e^{i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \\ e^{-i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} & \sin^2 \frac{\theta}{2} \end{pmatrix} + \begin{pmatrix} \sin^2 \frac{\theta}{2} & -e^{i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ -e^{-i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} & \cos^2 \frac{\theta}{2} \end{pmatrix} \\
 &= \begin{pmatrix} \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} & 0 \\ 0 & \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{I}_{2 \times 2}
 \end{aligned}$$

Now if we take $\hat{\mathbf{n}} = \hat{\mathbf{k}}$ then $\hat{\mathbf{n}} \cdot \hat{\mathbf{S}} = \hat{S}_z$. For this case we write $\theta = 0$. Now putting this in Eq. (1.168) we get :

$$|\lambda_+\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\lambda_-\rangle_z = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad (1.169)$$

This is exactly same as the solution we got in Eqs. (1.144) and (1.145). Now again if we take $\hat{\mathbf{n}} = \hat{\mathbf{i}}$ then $\hat{\mathbf{n}} \cdot \hat{\mathbf{S}} = \hat{S}_x$ and $\theta = \frac{\pi}{2}$ and $\phi = 0$. Which gives :

$$|\lambda_+\rangle_x = \begin{pmatrix} \cos \frac{\pi}{4} \\ \sin \frac{\pi}{4} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\lambda_-\rangle_x = \begin{pmatrix} \sin \frac{\pi}{4} \\ -\cos \frac{\pi}{4} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.170)$$

Similarly by taking $\hat{\mathbf{n}} = \hat{\mathbf{j}}$ gives $\theta = \frac{\pi}{2}$ and $\phi = \frac{\pi}{2}$. Which gives :

$$|\lambda_+\rangle_y = \begin{pmatrix} \cos \frac{\pi}{4} \\ e^{i\frac{\pi}{2}} \sin \frac{\pi}{4} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |\lambda_-\rangle_y = \begin{pmatrix} \sin \frac{\pi}{4} \\ -e^{i\frac{\pi}{2}} \cos \frac{\pi}{4} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (1.171)$$

Everything is fine. Again instead of finding $a_{\pm}$ we could have done it other way round by finding $b_{\pm}$ first. By Eq. (1.167) we can write :

$$a_{\pm} = \pm [\tan \frac{\theta}{2}]^{\mp 1} e^{-i\phi} b_{\pm} \quad (1.172)$$

And using normalization we get :

$$|b_{\pm}|^2 = \frac{1}{1 + [\tan \frac{\theta}{2}]^{\mp 2}}$$

Which gives us eigenstates as :

$$|\lambda_+\rangle = \begin{pmatrix} e^{-i\phi} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}, \quad |\lambda_-\rangle = \begin{pmatrix} -e^{-i\phi} \sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \end{pmatrix} \quad (1.173)$$

Now if we put $\hat{\mathbf{n}} = \hat{\mathbf{k}} \Rightarrow \theta = 0$.

$$|\lambda_+\rangle_z = \begin{pmatrix} e^{-i\phi} \\ 0 \end{pmatrix}, \quad |\lambda_-\rangle_z = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.174)$$

If we compare our results with Eqs. (1.144) and (1.145), then we will see we got a extra phase of $e^{-i\phi}$ and $e^{i\pi}$ in

$|\lambda_{\pm}\rangle_z$ respectively. Which is arbitrary anyway. So you might think how are we getting different states while proceeding via different route. But basically they are the same. The answer lies in the fact that, while calculating $a_{\pm}$ (similarly for $b_{\pm}$ which is done later) we could have added an extra phase factor that is arbitrary (say $e^{i\tilde{\phi}_{\pm}}$). Then Eq. (1.168) becomes :

$$|\lambda_{+}\rangle = \begin{pmatrix} e^{i\tilde{\phi}_{+}} \cos \frac{\theta}{2} \\ e^{i(\phi+\tilde{\phi}_{+})} \sin \frac{\theta}{2} \end{pmatrix}, \quad |\lambda_{-}\rangle = \begin{pmatrix} e^{i\tilde{\phi}_{-}} \sin \frac{\theta}{2} \\ -e^{i(\phi+\tilde{\phi}_{-})} \cos \frac{\theta}{2} \end{pmatrix} \quad (1.175)$$

Now taking $\tilde{\phi}_{+} = -\phi$ and $\tilde{\phi}_{-} = \pi - \phi$ we get back to Eq. (1.173). So what we have learned from here is that we can take the form given by Eq. (1.175) and put any value of $\tilde{\phi}_{\pm}$ as we want. They are all equivalent.

Problem 1.42

(a). Find the probability of measuring the “z” component of spin to be $\pm \frac{\hbar}{2}$ for the $\hat{\mathbf{n}} \cdot \hat{\mathbf{S}}$.

Hint : Write $|\lambda_{\pm}\rangle$ in terms of $|\uparrow\rangle$ and $|\downarrow\rangle$. i.e. say :

$$|\lambda_{\pm}\rangle = C_{\pm} |\uparrow\rangle + D_{\pm} |\downarrow\rangle \quad (1.176)$$

Then the corresponding probabilities will be $|C_{\pm}|^2$ and $|D_{\pm}|^2$. Alternatively what we can do the same by computing $|\langle\uparrow|\lambda_{\pm}\rangle|^2$ for getting the component as $\frac{\hbar}{2}$ and $|\langle\downarrow|\lambda_{\pm}\rangle|^2$ for getting the component as $-\frac{\hbar}{2}$ respectively.

(b). What about for the probabilities for “x” and “y” component of the spin.

Problem 1.43 “Spin $\frac{3}{2}$ ”

Consider $s = \frac{3}{2}$. Find out the matrix representation of the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z, \hat{\mathbf{S}}^2, \hat{S}_x^2, \hat{S}_y^2$ within the basis of $\hat{\mathbf{S}}^2$ and $\hat{S}_z$.

Example 1.15

For a spin $\frac{3}{2}$ particle, the Hamiltonian is given by :

$$\hat{H} = \frac{\epsilon_0}{\hbar^2} (\hat{S}_x^2 - \hat{S}_y^2) - \frac{\epsilon_0}{\hbar} \hat{S}_z \quad (1.177)$$

Find out the energy levels and also the eigenstates of the problem. Draw the energy spectrum also.

Ans: Using the form of the matrices found out in **Problem (1.43)** we can write the Hamiltonian in matrix form as follows :

$$\hat{H} = \frac{\epsilon_0}{2} \begin{pmatrix} -3 & 0 & 2\sqrt{3} & 0 \\ 0 & -1 & 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 & 1 & 0 \\ 0 & 2\sqrt{3} & 0 & 3 \end{pmatrix} \quad (1.178)$$

The eigenvalue equation becomes $\hat{H}|\psi\rangle = E|\psi\rangle$. Eigenvalues can be obtained by solving the characteristic equation $|\hat{H} - E\hat{I}| = 0$. Which is :

$$\begin{vmatrix} -\frac{3\epsilon_0}{2} - E & 0 & \sqrt{3}\epsilon_0 & 0 \\ 0 & -\frac{\epsilon_0}{2} - E & 0 & \sqrt{3}\epsilon_0 \\ \sqrt{3}\epsilon_0 & 0 & \frac{\epsilon_0}{2} - E & 0 \\ 0 & \sqrt{3}\epsilon_0 & 0 & \frac{3\epsilon_0}{2} - E \end{vmatrix} = 0 \quad (1.179)$$

Solving this we get the eigenvalues as :

$$E_1 = -\frac{5}{2}\epsilon_0, \quad E_2 = -\frac{3}{2}\epsilon_0, \quad E_3 = +\frac{3}{2}\epsilon_0, \quad E_4 = +\frac{5}{2}\epsilon_0 \quad (1.180)$$

Now for the eigenstate corresponding to E_1

Write the eigenstate as $|\psi_1\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$. Eigenvalue equation is $\hat{H}|\psi_1\rangle = E_1|\psi_1\rangle$. Which gives :

$$\frac{\epsilon_0}{2} \begin{pmatrix} -3 & 0 & 2\sqrt{3} & 0 \\ 0 & -1 & 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 & 1 & 0 \\ 0 & 2\sqrt{3} & 0 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = -\frac{5}{2}\epsilon_0 \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

Which gives :

$$\Rightarrow \begin{cases} -3a + 2\sqrt{3}c = -5a \rightarrow a = -\sqrt{3}c \\ -b + 2\sqrt{3}d = -5b \rightarrow 4b = -2\sqrt{3}d \\ -2\sqrt{3}a + c = -5c \rightarrow a = -\sqrt{3}c \\ 2\sqrt{3}b + 3d = -5d \rightarrow \sqrt{3}b = -4d \end{cases}$$

We immediately get $b = d = 0$ and normalization gives $a^2 + c^2 = 1$ or $c = 1/2$ and $a = -\sqrt{3}/2$. So finally we have the state as : $|\psi_1\rangle = \frac{1}{2} \begin{pmatrix} -\sqrt{3} \\ 0 \\ 1 \\ 0 \end{pmatrix}$. Similarly other states are :

$$|\psi_2\rangle = \frac{1}{2} \begin{pmatrix} 0 \\ -\sqrt{3} \\ 0 \\ 1 \end{pmatrix}, \quad |\psi_3\rangle = \frac{1}{\sqrt{12}} \begin{pmatrix} \sqrt{3} \\ 0 \\ 3 \\ 0 \end{pmatrix}, \quad |\psi_4\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ \sqrt{3} \\ 0 \end{pmatrix}$$

Now it is your job to check this forms a complete basis. i.e. $\sum_{n=1}^4 |n\rangle\langle n| = \hat{I}_{4 \times 4}$

Problem 1.44

In continuation of **Example (1.15)**, if the system was initially (at $t = 0$) in state $|\psi_0\rangle = \frac{1}{\sqrt{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$ then find the state of the system at a later time " $t > 0$ ", i.e. "time evolution".

Hint : Write $|\psi_0\rangle$ in terms of the basis states, i.e.

$$|\psi_0\rangle = \sum_{n=1}^4 C_n |n\rangle \quad (1.181)$$

You can find the coefficient $C_n = \langle n|\psi_0\rangle$. Then we immediately get :

$$|\psi(t)\rangle = \sum_{n=1}^4 C_n |n\rangle e^{-iE_n t/\hbar} \quad (1.182)$$

Example 1.16

Find out the energy levels of a spin " $\frac{5}{2}$ " particle whose Hamiltonian is given by :

$$\hat{H} = \frac{\epsilon_0}{\hbar^2} (\hat{S}_x^2 + \hat{S}_y^2) + \frac{\epsilon_0}{\hbar} \hat{S}_z \quad (1.183)$$

Ans: Rewriting the hamiltonian in terms of $\hat{\mathbf{S}}^2$, $\hat{S}_z^2$ and $\hat{S}_z$ we have :

$$\hat{H} = \frac{\epsilon_0}{\hbar^2} \hat{\mathbf{S}}^2 - \frac{\epsilon_0}{\hbar^2} \hat{S}_z^2 + \frac{\epsilon_0}{\hbar} \hat{S}_z \quad (1.184)$$

The eigenvalue equation becomes $\hat{H}|\psi\rangle = E|\psi\rangle$ and as this hamiltonian commutes with $\hat{\mathbf{S}}^2$ and $\hat{S}_z$, i.e.

$$[\hat{H}, \hat{\mathbf{S}}^2] = 0 \quad (1.185a)$$

$$[\hat{H}, \hat{S}_z] = 0 \quad (1.185b)$$

The eigenstates $|s, m_s\rangle$ is also a eigenstate of the given Hamiltonian with energy eigenvalue :

$$\begin{aligned} E_{s, m_s} &= \frac{\epsilon_0}{\hbar^2} s(s+1)\hbar^2 - \frac{\epsilon_0}{\hbar^2} m_s^2\hbar^2 + \frac{\epsilon_0}{\hbar} m_s\hbar \\ &= \epsilon_0 [s(s+1) - m_s^2 + m_s] \\ &= \epsilon_0 [s(s+1) - m_s(m_s - 1)] \end{aligned}$$

Where $s = \frac{5}{2}$ and $m_s = -\frac{5}{2}, -\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}, +\frac{5}{2}$.

Problem 1.45

For the Hamiltonian given in **Example (1.16)**, are the energy levels degenerate ? Draw the “Energy spectrum” of this system.

Problem 1.46

Similar to **Example (1.15)** take a $\frac{3}{2}$ particle whose Hamiltonian is given by :

$$\hat{H} = \frac{\epsilon_0}{\hbar^2} (\hat{S}_x^2 - \hat{S}_y^2) - \frac{\epsilon_0}{\hbar^2} \hat{S}_z^2 \quad (1.186)$$

- (a) By using results obtained in **Problem (1.43)** obtain the matrix form of this Hamiltonian.
- (b) Find out the energy eigenvalues and corresponding eigenstates.
- (c) Verify energy eigenvalues are doubly degenerate.

Hint: $\hat{H}$ is a 4×4 matrix. So it has 4 eigenvalues. If “energy levels” are doubly degenerate say the energy eigenvalues are “ E_1, E_2 ” and again “ E_1, E_2 ”. If the eigenvectors are $|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle$ and if $\langle\psi_1|\psi_3\rangle = \langle\psi_2|\psi_4\rangle = 0$ then we will know they are orthogonal states. Also check that they form “complete basis”.

Problem 1.47 “More generic Problem” **

The spin angular momentum matrix construction is exactly the same. Use the same code that you had written in **Problem (1.18)** to construct $\hat{S}_x, \hat{S}_y, \hat{S}_z, \hat{\mathbf{S}}^2, \hat{S}_\pm$ for a given s . You can work in natural units ($\hbar = 1$). Otherwise if we put the value of $\hbar$, “computationally” the matrix elements would be too small.

Now consider a generic Hamiltonian operator in terms of spin operators :

$$\hat{H} = \frac{1}{\hbar^2} \sum_{ij} \alpha_{ij} \hat{S}_i \hat{S}_j + \frac{1}{\hbar} \sum_k \beta_k \hat{S}_k \quad (1.187)$$

The factors of $\hbar$ are taken so that the coefficients α_{ij}, β_k ⁴⁰ have units of energy.

It is evident that this is a more general Hamiltonian as we have seen **Example (1.15)**, **Example (1.16)**, **Problem (1.46)**. Comparing this with

1. **Example (1.15)** we get $\alpha_{xx} = \epsilon_0, \alpha_{yy} = -\epsilon_0, \beta_z = -\epsilon_0$.

2. **Example (1.16)** we get $\alpha_{xx} = \epsilon_0, \alpha_{yy} = \epsilon_0, \beta_z = \epsilon_0$.

3. **Problem (1.46)** we get $\alpha_{xx} = \epsilon_0, \alpha_{yy} = -\epsilon_0, \alpha_{zz} = -\epsilon_0$.

You can take $\hbar = 1$, or any ways the factor will cancel arising from spin operators. Construct α matrix using the given α_{ij} 's and β vector explicitly. Afterwards using the code that you had written to compute matrix representation of the operators and then construct $\hat{H}$ matrix numerically. Then solve for eigenvalue and eigenvectors. Make sure to check that the eigenvectors are “orthonormal” and they form a “complete basis”. Also compare with the results obtained in **Example (1.15)**, **Example (1.16)**, **Problem (1.46)**.

⁴⁰These must not be real all the time, All we need is the Hamiltonian to be hermitian.

Proof of generic commutation

If $\vec{a} \times \vec{b} = \vec{c}$, then $c_i = (\vec{a} \times \vec{b})_i = \epsilon_{ijk} a_j b_k$. Using this we can write $\hat{L}_i = \epsilon_{ijk} \hat{r}_j \hat{p}_k$.

Now considering the commutator :

$$\begin{aligned} [\hat{L}_i, \hat{L}_j] &= [\epsilon_{i\alpha\beta} \hat{r}_\alpha \hat{p}_\beta, \epsilon_{j\gamma\delta} \hat{r}_\gamma \hat{p}_\delta] \\ &= \epsilon_{i\alpha\beta} \epsilon_{j\gamma\delta} [\hat{r}_\alpha \hat{p}_\beta, \hat{r}_\gamma \hat{p}_\delta] \end{aligned}$$

Now using the commutations :

$$\begin{aligned} [\hat{A}\hat{B}, \hat{C}\hat{D}] &= \hat{A}[\hat{B}, \hat{C}]\hat{D} + [\hat{A}, \hat{C}]\hat{B}\hat{D} + \hat{C}\hat{A}[\hat{B}, \hat{D}] + \hat{C}[\hat{A}, \hat{D}]\hat{B} \\ [\hat{r}_\alpha, \hat{r}_\gamma] &= 0 \\ [\hat{p}_\beta, \hat{p}_\delta] &= 0 \\ [\hat{r}_\rho, \hat{p}_\lambda] &= i\hbar\delta_{\rho\lambda} \end{aligned}$$

We evaluate :

$$\begin{aligned} [\hat{r}_\alpha \hat{p}_\beta, \hat{r}_\gamma \hat{p}_\delta] &= \hat{r}_\alpha [\hat{p}_\beta, \hat{r}_\gamma] \hat{p}_\delta + \hat{r}_\gamma [\hat{r}_\alpha, \hat{p}_\delta] \hat{p}_\beta \\ &= i\hbar(\hat{r}_\gamma \hat{p}_\beta \delta_{\alpha\delta} - \hat{r}_\alpha \hat{p}_\delta \delta_{\beta\gamma}) \end{aligned}$$

So finally we have :

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{i\alpha\beta} \epsilon_{j\gamma\delta} (\hat{r}_\gamma \hat{p}_\beta \delta_{\alpha\delta} - \hat{r}_\alpha \hat{p}_\delta \delta_{\beta\gamma}) \quad (\text{A.1})$$

First term :

$$\begin{aligned} T_1 &= i\hbar \epsilon_{i\alpha\beta} \epsilon_{j\gamma\delta} \hat{r}_\gamma \hat{p}_\beta \delta_{\alpha\delta} \\ &= i\hbar \epsilon_{i\alpha\beta} \epsilon_{j\gamma\alpha} \hat{r}_\gamma \hat{p}_\beta \quad (\text{using } \delta_{\alpha\delta}) \\ &= i\hbar \epsilon_{i\beta\alpha} \epsilon_{\alpha\gamma j} \hat{r}_\gamma \hat{p}_\beta \\ &= i\hbar (\delta_{ij} \delta_{\beta\gamma} - \delta_{ij} \delta_{\beta\gamma}) \hat{r}_\gamma \hat{p}_\beta \\ &= i\hbar (\hat{r}_i \hat{p}_j - \delta_{ij} \hat{r}_\gamma \hat{p}_\gamma) \end{aligned}$$

Where we have used the identity of “levi-civita” :

$$\epsilon_{abc} \epsilon_{cde} = \delta_{ad} \delta_{be} - \delta_{ae} \delta_{bd} \quad (\text{A.2})$$

Second term :

$$\begin{aligned}
T_2 &= -i\hbar \epsilon_{i\alpha\beta} \epsilon_{j\gamma\delta} \hat{r}_\alpha \hat{p}_\delta \delta_{\beta\gamma} \\
&= -i\hbar \epsilon_{i\alpha\beta} \epsilon_{j\beta\delta} \hat{r}_\alpha \hat{p}_\delta \quad (\text{using } \delta_{\beta\gamma}) \\
&= -i\hbar \epsilon_{i\alpha\beta} \epsilon_{\beta\delta j} \hat{r}_\alpha \hat{p}_\delta \\
&= -i\hbar (\delta_{i\delta} \delta_{\alpha j} - \delta_{ij} \delta_{\alpha\delta}) \hat{r}_\alpha \hat{p}_\delta \\
&= -i\hbar (\hat{r}_j \hat{p}_i - \delta_{ij} \hat{r}_\alpha \hat{p}_\alpha)
\end{aligned}$$

Finally we have :

$$\begin{aligned}
[\hat{L}_i, \hat{L}_j] &= T_1 + T_2 \\
&= i\hbar (\hat{r}_i \hat{p}_j - \delta_{ij} \hat{r}_\gamma \hat{p}_\gamma) - i\hbar (\hat{r}_j \hat{p}_i - \delta_{ij} \hat{r}_\alpha \hat{p}_\alpha) \\
&= i\hbar (\hat{r}_i \hat{p}_j - \hat{r}_j \hat{p}_i)
\end{aligned}$$

Now by computing :

$$\begin{aligned}
\epsilon_{ijk} \hat{L}_k &= \epsilon_{ijk} \epsilon_{k\mu\nu} \hat{r}_\mu \hat{p}_\nu \\
&= (\delta_{i\mu} \delta_{j\nu} - \delta_{i\nu} \delta_{j\mu}) \hat{r}_\mu \hat{p}_\nu \\
&= \hat{r}_i \hat{p}_j - \hat{r}_j \hat{p}_i
\end{aligned}$$

We can finally write :

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k \quad (\text{A.3})$$

Evaluation of Integral

$$\begin{aligned}
 I_\ell &= \int_0^\pi \sin \theta \, d\theta (\sin \theta)^{2\ell} \\
 &= \int_0^\pi \sin \theta \, d\theta (\sin^2 \theta)^\ell \\
 &= \int_0^\pi \sin \theta \, d\theta (1 - \cos^2 \theta)^\ell
 \end{aligned}$$

Take $t = \cos \theta$ which gives $dt = -\sin \theta \, d\theta$ and the limit of t is from $+1$ to -1 . So the integral becomes :

$$\begin{aligned}
 I_\ell &= - \int_{+1}^{-1} dt (1 - t^2)^\ell \\
 &= \int_{-1}^{+1} dt (1 - t^2)^\ell \\
 &= 2 \int_0^1 dt (1 - t^2)^\ell \\
 &= \int_0^1 d(t^2) (t^2)^{-\frac{1}{2}} (1 - t^2)^\ell \\
 &= \int_0^1 d(t^2) (t^2)^{\frac{1}{2}-1} (1 - t^2)^{\ell+1-1} \\
 &= \beta\left(\frac{1}{2}, \ell + 1\right) \\
 &= \frac{\Gamma(\frac{1}{2}) \times \Gamma(\ell + 1)}{\Gamma(\frac{1}{2} + \ell + 1)} \\
 &= \frac{2^{2\ell+1} (\ell!)^2}{(2\ell + 1)!} \text{¹}
 \end{aligned}$$

¹Using the identity $\Gamma(n + 1) = n \Gamma(n)$

$$\begin{aligned}
 \Gamma\left(\ell + \frac{1}{2} + 1\right) &= \left(\ell + \frac{1}{2}\right) \Gamma\left(\ell + \frac{1}{2}\right) \\
 &= \left(\ell + \frac{1}{2}\right) \left(\ell - \frac{1}{2}\right) \Gamma\left(\ell - \frac{1}{2}\right) \\
 &= \left(\ell + \frac{1}{2}\right) \left(\ell - \frac{1}{2}\right) \left(\ell - \frac{3}{2}\right) \Gamma\left(\ell - \frac{3}{2}\right) \\
 &\vdots \\
 &\vdots
 \end{aligned}$$

$$\begin{aligned}
& \vdots \\
& \vdots \\
& = \underbrace{\left(\ell + \frac{1}{2}\right)\left(\ell - \frac{1}{2}\right)\left(\ell - \frac{3}{2}\right) \cdots \frac{3}{2} \frac{1}{2}}_{(\ell+1) \text{ terms}} \Gamma\left(\frac{1}{2}\right) \\
& = \frac{(2\ell+1)(2\ell-1)(2\ell-3) \cdots 3 \cdot 1}{2^{\ell+1}} \Gamma\left(\frac{1}{2}\right) \\
& = \frac{(2\ell+1)(2\ell)(2\ell-1)(2\ell-2)(2\ell-3) \cdots 3 \cdot 2 \cdot 1}{2^{\ell+1} \times \underbrace{(2\ell)(2\ell-2) \cdots 2}_{\ell \text{ terms}}} \Gamma\left(\frac{1}{2}\right) \\
& = \frac{(2\ell+1)!}{2^{2\ell+1} \times \ell!} \Gamma\left(\frac{1}{2}\right)
\end{aligned}$$